TRIPLE QUANTUM SCHUBERT CALCULUS

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ABSTRACT. We study the structure constants $c_{u,v}^{w,d}(y;z)$ that arise when a product of two quantum double Schubert polynomials in *different* sets of equivariant parameters, $\widetilde{\mathfrak{S}}_u(x;y)\,\widetilde{\mathfrak{S}}_v(x;z)$, is expanded in the basis $\{\widetilde{\mathfrak{S}}_w(x;y)\}$. We call them triple quantum Schubert structure constants; they specialize to the equivariant quantum Littlewood–Richardson coefficients when $z = y$ and to the ordinary triple Schubert structure constants when $q = 0$. Adapting an argument of Xiao and Yue, we show that they are polynomials in the differences $y_i - z_j$ with nonnegative integer coefficients. We prove a Pieri formula for multiplication by a quantum factorial elementary polynomial, indexed by a quantum analogue of Sottile's relation, in which each coefficient is a monomial in q times a product of differences $y_a - z_1$; the proof is a self-contained analysis of chains in the quantum Bruhat graph. We extend the Peterson–Woodward comparison formula and the quantum-to-classical reduction of Huang and Li to the triple constants, thereby reducing the computation of the parabolic triple quantum structure constants to the complete flag case. Finally we describe the algorithms implemented in the package `schubmult` for computing these constants, and give an interpretation of the coefficients in terms of the Fomin–Gelfand–Postnikov difference operators.

Keywords: quantum double Schubert polynomial, structure constant, Littlewood-Richardson rule, Pieri formula, quantum Bruhat graph, Peterson–Woodward comparison formula

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1. INTRODUCTION

Let $\mathfrak{S}_w(x;y)$, $w \in S_\infty$, denote the double Schubert polynomials of Lascoux and Schützenberger, which represent the torus-equivariant Schubert classes of the complete flag variety, and let $\widetilde{\mathfrak{S}}_w(x;y)$ denote the quantum double Schubert polynomials of Kirillov and Maeno [9], which represent the Schubert classes in torus-equivariant quantum cohomology [10]. The structure constants of the equivariant quantum cohomology ring are the coefficients in

$$\widetilde{\mathfrak{S}}_u(x;y)\,\widetilde{\mathfrak{S}}_v(x;y) = \sum_{w,d} q^d\, c_{u,v}^{w,d}(y;y)\;\widetilde{\mathfrak{S}}_w(x;y),$$

and Mihalcea [12] proved that they are polynomials in the negative simple roots $y_{i+1} - y_i$ with nonnegative integer coefficients.

In this paper we study the coefficients obtained when the two factors carry *different* equivariant parameters:

$$\widetilde{\mathfrak{S}}_u(x;y)\,\widetilde{\mathfrak{S}}_v(x;z) = \sum_{w,d} q^d\, c_{u,v}^{w,d}(y;z)\;\widetilde{\mathfrak{S}}_w(x;y).$$

We call the $c_{u,v}^{w,d}(y;z)$ the *triple quantum Schubert structure constants*. At $z = y$ they recover the equivariant quantum structure constants, and at $q = 0$ they recover the triple Schubert structure constants $c_{u,v}^{w}(y;z)$ of [16], which were recently shown by Gao and Xiong [6] to be polynomials in the differences $y_i - z_j$ with nonnegative integer coefficients. The triple constants carry strictly more information than the equivariant ones: the equivariant quantum constant $c_{u,v}^{w,d}(y;y)$ is a specialization of $c_{u,v}^{w,d}(y;z)$, while conversely $c_{u,v}^{w,d}(y;z)$ is recovered from the constants $c_{u,v_2}^{w,d}(y;y)$, v_2 ranging over the right factors of the reduced factorizations $v = v_1 \cdot v_2$, by the quantum Cauchy identity (Lemma 2.1).

Our main results are the following.

(1) *Positivity* (Theorem 2.3). The constants $c_{u,v}^{w,d}(y;z)$ lie in $\mathbb{Z}_{\geq 0}[y_i - z_j]$. The proof, which we reproduce from the manuscript of Xiao and Yue [21] with their permission, combines the geometric argument of Mihalcea with the Graham-positivity theorem of Gao and Xiong for B^--stable effective cycles.

(2) *A quantum Pieri formula* (Theorem 3.4). Let $\widetilde{E}_{k,k}(x;z) = \widetilde{\mathfrak{S}}_{s_1 s_2 \cdots s_k}(x;z)$ be the quantum factorial elementary polynomial. We prove

$$\widetilde{\mathfrak{S}}_u(x;y)\widetilde{E}_{k,k}(x;z) = \sum_{u \overset{k}{\rightsquigarrow} w} q_k(u,w) \prod_{a \in \mathrm{fix}_k(u,w)} (y_a - z_1)\ \widetilde{\mathfrak{S}}_w(x;y),$$

where $\overset{k}{\rightsquigarrow}$ is a quantum analogue of the relation introduced by Sottile [19] for the classical Pieri rule, $q_k(u,w)$ is a monomial in the q_i recording the length-decreasing steps of a chain from u to w in the quantum Bruhat graph, and $\mathrm{fix}_k(u,w)$ is the set of values fixed by the chain in the first k positions. This is an explicit instance of the positivity theorem, and it is the engine of the multiplication algorithm. The proof occupies Section 6; it proceeds by induction on k via the single-variable Monk formula, and rests on an intrinsic description of the chains (every cycle of $u^{-1}w$ has exactly one element exceeding k, and a single length identity) which makes the term-by-term cancellations tractable. In the non-equivariant case the quantum Pieri formula for complete flag manifolds is due to Ciocan-Fontanine [3] and Postnikov [15].

(3) *Parabolic reduction* (Proposition 2.10). The Peterson–Woodward comparison formula, in the form proved by Huang and Li [7] for equivariant quantum cohomology, extends verbatim to the triple constants. Consequently the triple structure constants of the parabolic quantum double Schubert polynomials of Lam and Shimozono are equal to triple constants of the complete flag variety, and in particular they are also Graham-positive (Corollary 2.12).

(4) *Algorithms and operators* (Sections 4 and 5). We describe the algorithms implemented in the package `schubmult` [18] for computing products of (quantum) double Schubert polynomials, which combine the Pieri formula on one factor with a signed expansion of the other factor into products of factorial elementary polynomials. We also show that the Fomin–Gelfand–Postnikov quantization operators [5] package the triple constants: the coefficient of ∂^u in $\widetilde{\mathfrak{S}}_v(\chi^{(n)};z)$ is $\sum_d q^d c^{1,d}_{u,v}(x;z)$. Finally, we extend the quantum-to-classical reduction of Huang and Li to the triple constants.

The paper is organized as follows. Section 2 fixes notation, proves the positivity theorem and treats the parabolic case. Section 3 states the Pieri formula, whose proof is given in Section 6. Section 4 describes the algorithms and the operator interpretation, Section 5 the quantum-to-classical reduction, Section 7 the iterated Pieri formula, and Section 8 the Peterson–Woodward comparison formula.

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2. Definitions and statement of results

2.1. **Quantum double Schubert polynomials and triple quantum Schubert structure constants.** We fix $n \geq 1$ and write

$$x = (x_1, \ldots, x_n), \qquad y = (y_1, \ldots, y_n), \qquad q = (q_1, \ldots, q_{n-1}).$$

For $1 \leq a < b \leq n$, set

$$q_{ab} := q_a q_{a+1} \cdots q_{b-1},$$

and use the cohomological grading $\deg q_i = 2$ (equivalently $\deg q_{ab} = 2(b-a)$).

Let s_i act on the y-variables by swapping y_i, y_{i+1}, and define the divided difference operators

$$\partial_i^{(y)} f := \frac{f - s_i f}{y_i - y_{i+1}} \qquad (1 \leq i \leq n-1).$$

For $w \in S_n$, if $w = s_{i_1} \cdots s_{i_r}$ is any reduced word, set

$$\partial_w^{(y)} := \partial_{i_1}^{(y)} \cdots \partial_{i_r}^{(y)}.$$

Following Kirillov–Maeno [9], define the quantum elementary polynomials $e_j^{(q)}(x_1, \ldots, x_k)$ by

$$\Delta_k(t \mid x_1, \ldots, x_k) := \sum_{j=0}^{k} t^{k-j} e_j^{(q)}(x_1, \ldots, x_k),$$

where

$$\Delta_k(t \mid x_1, \ldots, x_k) = \det \begin{pmatrix} x_1 + t & q_1 & 0 & \cdots & 0 \\ -1 & x_2 + t & q_2 & \ddots & \vdots \\ 0 & -1 & x_3 + t & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & q_{k-1} \\ 0 & \cdots & 0 & -1 & x_k + t \end{pmatrix}.$$

Let $w_0 \in S_n$ be the longest permutation. The top quantum double Schubert polynomial is

$$\widetilde{\mathfrak{S}}_{w_0}(x; y) := \prod_{i=1}^{n-1} \Delta_i(-y_{n-i} \mid x_1, \ldots, x_i),$$

and for $w \in S_n$ we define

$$\widetilde{\mathfrak{S}}_w(x; y) := (-1)^{\ell(w_0)-\ell(w)} \partial^{(y)}_{ww_0} \widetilde{\mathfrak{S}}_{w_0}(x; y).$$

Its specialization

$$\widetilde{\mathfrak{S}}_w(x) := \widetilde{\mathfrak{S}}_w(x; 0)$$

is the quantum Schubert polynomial. In particular, at $q = 0$ these recover the ordinary double and single Schubert polynomials.

Given their stability property, we may let the number of variables be infinite, indexing the polynomials by S_∞. As a module over $\mathbb{Z}[q][y]$, the quantum double Schubert polynomials form a basis of $\mathbb{Z}[q][x, y]$, and we may write

$$\widetilde{\mathfrak{S}}_u(x; y)\, \widetilde{\mathfrak{S}}_v(x; z) = \sum_{w,d} q^d c^{w,d}_{u,v}(y; z)\, \widetilde{\mathfrak{S}}_w(x; y)$$

where $d = (d_1, \ldots, d_{n-1})$ is a weak composition and $q^d = q_1^{d_1} \cdots q_{n-1}^{d_{n-1}}$. The coefficients $c^{w,d}_{u,v}(y; z)$ are the *triple quantum Schubert structure constants*, which are polynomials in y and z with integer coefficients. This product respects the grading where for each i, $\deg y_i = \deg z_i = 1$ and $\deg q_i = 2$. In particular, if $c^{w,d}_{u,v}(y; z) \neq 0$, then $\deg c^{w,d}_{u,v}(y; z) = \ell(u) + \ell(v) - \ell(w) + 2\sum_i d_i$. Unlike the usual triple Schubert structure constants, they do not exhibit triangularity in Bruhat order, and indeed we may have $\ell(w) < \ell(u) + \ell(v)$. There is an ordering in which the quantum double Schubert polynomials are triangular, but we will not need this in full generality.

Throughout, a factorization $w = v_1 \cdot v_2$ written with a dot is understood to be *reduced*, i.e. $\ell(w) = \ell(v_1) + \ell(v_2)$. The following identity will be used repeatedly to pass between the triple constants and the equivariant quantum structure constants $c^{w,d}_{u,v}(y; y)$. Setting $y = 0$ in it recovers the expansion $\widetilde{\mathfrak{S}}_w(x; z) = \sum_{v_1 \cdot v_2 = w} \mathfrak{S}_{v_1}(0; z)\, \widetilde{\mathfrak{S}}_{v_2}(x)$ of Kirillov–Maeno [9] in terms of quantum Schubert polynomials.

Lemma 2.1 (Quantum Cauchy identity)**.** *For all $w \in S_\infty$,*

$$\widetilde{\mathfrak{S}}_w(x; z) = \sum_{v_1 \cdot v_2 = w} \mathfrak{S}_{v_1}(y; z)\ \widetilde{\mathfrak{S}}_{v_2}(x; y).$$

Consequently, for all u, v, w and d,

$$c^{w,d}_{u,v}(y; z) = \sum_{v_1 \cdot v_2 = v} c^{w,d}_{u,v_2}(y; y)\ \mathfrak{S}_{v_1}(y; z).$$

Proof. Let θ denote the quantization map of Fomin, Gelfand and Postnikov [5], the $\mathbb{Z}[q]$-linear map on polynomials in x which sends each standard elementary monomial $\prod_j e_{i_j}(x_1, \ldots, x_j)$ to $\prod_j e^{(q)}_{i_j}(x_1, \ldots, x_j)$, extended linearly over any further coefficient variables. Since the divided differences in the y-variables commute with θ, the definition of $\widetilde{\mathfrak{S}}_w(x; y)$ by y-divided differences gives $\widetilde{\mathfrak{S}}_w(x; y) = \theta(\mathfrak{S}_w(x; y))$ [10, (10)]. Now apply θ, which is $\mathbb{Z}[y, z]$-linear, to the classical Cauchy-type expansion $\mathfrak{S}_w(x; z) = \sum_{v_1 \cdot v_2 = w} \mathfrak{S}_{v_1}(y; z)\, \mathfrak{S}_{v_2}(x; y)$ [11, (6.3)]: the left side becomes $\widetilde{\mathfrak{S}}_w(x; z)$ and the right side becomes $\sum_{v_1 \cdot v_2 = w} \mathfrak{S}_{v_1}(y; z)\, \widetilde{\mathfrak{S}}_{v_2}(x; y)$. For the second assertion, multiply the first identity (with $w = v$) by $\widetilde{\mathfrak{S}}_u(x; y)$ and expand each $\widetilde{\mathfrak{S}}_u(x; y)\, \widetilde{\mathfrak{S}}_{v_2}(x; y)$ in the basis $\{\widetilde{\mathfrak{S}}_w(x; y)\}$ with coefficients $q^d c^{w,d}_{u,v_2}(y; y)$. □

Remark 2.2 ($\overset{k}{\rightsquigarrow}$ and the quantum k-Bruhat order). The *quantum k-Bruhat order* was introduced by Benedetti, Bergeron, Colmenarejo, Saliola, and Sottile [1] and studied further by Colmenarejo and Mayers [4]. That order $\preceq_k^{\mathbf{q}}$ on $S_\infty[\mathbf{q}]$ is generated by cover relations

$$u' \lessdot_k^{\mathbf{q}} \mathbf{q}^\gamma u' t_{ab}, \qquad a \le k < b,$$

each of exactly two kinds: a *classical* cover, with $\gamma = 0$ and $\ell(u't_{ab}) = \ell(u') + 1$, or a *quantum* cover, with $\ell(u') + 1 = \ell(\mathbf{q}^\gamma u' t_{ab}) = \ell(u't_{ab}) + 2(b-a)$, that is $\ell(u't_{ab}) = \ell(u') - 2(b-a) + 1$ and $\gamma = e_a + \cdots + e_{b-1}$. These are exactly the two alternatives of condition (4) of Definition 3.1. Consequently every sequence $t_{a_1 b_1}, \ldots, t_{a_p b_p}$ witnessing $u \overset{k}{\rightsquigarrow} w$ is a saturated chain from u to $\mathbf{q}^\alpha w$ in the quantum k-Bruhat order, where $\mathbf{q}^\alpha$ is the monomial recorded by $q_k(u, w)$; in particular $u \overset{k}{\rightsquigarrow} w$ implies $u \preceq_k^{\mathbf{q}} \mathbf{q}^\alpha w$.

The relation $\overset{k}{\rightsquigarrow}$ is *not* the full order $\preceq_k^{\mathbf{q}}$, however. Definition 3.1 imposes the additional requirements that the lower indices $a_1, \ldots, a_p$ be pairwise distinct, so that $\overset{k}{\rightsquigarrow}$ admits only those saturated chains of a special form. Thus $\overset{k}{\rightsquigarrow}$ is strictly larger than the covering relation $\lessdot_k^{\mathbf{q}}$ but strictly smaller than $\preceq_k^{\mathbf{q}}$: a general chain in the quantum k-Bruhat order may reuse a lower index a_i, and such chains do not contribute to the Pieri expansion. This is the quantum counterpart of the classical situation, where the Pieri relation of [19, 16] is likewise the restriction of the k-Bruhat order to chains with distinct lower indices, and it is exactly this restriction that makes the Pieri coefficients well defined.

What we use below is only that each individual step of a $\overset{k}{\rightsquigarrow}$ chain is a cover, and hence is governed by the following length-free description of the covers [1, Proposition 3.1] (see also [4, Proposition 4.2]): with $i \le k < j$, the step $u' \lessdot_k^{\mathbf{q}} \mathbf{q}^\gamma u' t_{ij}$ is a classical cover exactly when $u'(i) < u'(j)$ and no l with $i < l < j$ satisfies $u'(i) < u'(l) < u'(j)$, and it is a quantum cover exactly when $u'(i) > u'(j)$ and every l with $i < l < j$ satisfies $u'(j) < u'(l) < u'(i)$. This characterization is an immediate consequence of the standard formula for the length of $u't_{ij}$ (Lemma 6.2 below), and it is the form in which the covers are used throughout Section 6.

For $d = (0, \ldots, 0)$, this is the ordinary triple Schubert calculus structure constant, and it was proved in [6] that these have the positivity property that they are polynomials in the differences $y_i - z_j$ with nonnegative integer coefficients. We will refer to this as *triple Schubert positivity*. Our first main result is that it persists in the quantum setting.

Theorem 2.3 (Triple quantum Schubert positivity)**.** *For all u, v, w and d, the structure constants $c_{u,v}^{w,d}(y; z)$ are polynomials in the differences $y_i - z_j$ with nonnegative integer coefficients.*

Proof. The proof below is adapted from the manuscript of Tongren Xiao and Yihua Yue [21], with their permission. Their product is written with the output basis carrying the parameters of the second factor. Commutativity of multiplication therefore identifies their coefficient $C_{v,u}^{w,d}(z, y)$ with $c_{u,v}^{w,d}(y; z)$. It is enough to prove that $C_{u,v}^{w,d}(y, t)$ is in $\mathbb{Z}_{\ge 0}[t_i - y_j]$.

Fix $u, v, w \in S_n$, increasing n if necessary so that all permutations and parameters occurring in the coefficient under consideration are included. Write $\mathbf{a} = (t_1, \ldots, t_n, y_1, \ldots, y_n)$, and let $\tau \in S_{2n}$ interchange the two blocks, so that $\tau\mathbf{a} = (y_1, \ldots, y_n, t_1, \ldots, t_n)$. Regard the product expansion in S_{2n}:

$$\widetilde{\mathfrak{S}}_u(x; \tau\mathbf{a})\, \widetilde{\mathfrak{S}}_v(x; \mathbf{a}) = \sum_{w \in S_{2n}} C_{u,v}^{w,\tau}(\mathbf{a}; q)\, \widetilde{\mathfrak{S}}_w(x; \mathbf{a}). \tag{2.1}$$

We recall the geometric interpretation of these coefficients. Let $e_i(x)$ and $E_i(x, q)$ denote, respectively, the elementary symmetric polynomials and their quantum analogues in $2n$ variables, and set

$$J_{\mathbf{a}}^{2n} = \langle e_i(x) - e_i(\mathbf{a}) : 1 \le i \le 2n \rangle, \qquad J_{q\mathbf{a}}^{2n} = \langle E_i(x, q) - e_i(\mathbf{a}) : 1 \le i \le 2n \rangle.$$

The Fomin–Gelfand–Postnikov deformation map [5], extended $\mathbb{Z}[\mathbf{a}]$-linearly, induces

$$\bar{\theta} : \mathbb{Z}[x, \mathbf{a}]/J_{\mathbf{a}}^{2n} \longrightarrow \mathbb{Z}[x, \mathbf{a}, q]/J_{q\mathbf{a}}^{2n}.$$

The classical Borel presentation identifies the source with $H_T^*(X)$, and Kim's presentation [8] identifies the target with $QH_T^*(X)$, where $X = \mathrm{Fl}_{2n}$. Under the Lam–Shimozono identification [10] of quantum double Schubert polynomials with quantum Schubert classes, (2.1) is the quantum equivariant Littlewood–Richardson expansion for the classes obtained by applying $\bar{\theta}$ to the first factor. Consequently,

$$C_{u,v}^{w,\tau}(\mathbf{a}; q) = \sum_{d \ge 0} q^d (\pi_{M_d})_*^T \left(\mathrm{ev}_1^*[\tau X_u]^T \ \mathrm{ev}_2^*[X_v]^T \ \mathrm{ev}_3^*[Y_w]^T \right). \tag{2.2}$$

Here $M_d = \overline{M}_{0,3}(X,d)$, $\mathrm{ev}_i : M_d \to X$ are the evaluation maps, $\pi_{M_d} : M_d \to \mathrm{pt}$ is the structure map, and X_r and Y_r denote the Schubert and opposite Schubert varieties indexed by r.

For each effective degree d, put

$$F = (\mathrm{ev}_1, \mathrm{ev}_2) : M_d \longrightarrow X \times X, \qquad I^d_{u,v} = F^{-1}(\tau X_u \times X_v), \qquad \Gamma^d_{u,v} = (\mathrm{ev}_3)_*[I^d_{u,v}].$$

The incidence scheme $I^d_{u,v}$ is reduced and locally irreducible, and all its irreducible components have codimension $\ell(u)+\ell(v)$. Indeed, the moduli space M_d is irreducible by Thomsen's theorem [20], and Mihalcea's incidence lemma [12, Lemma 6.3] applies after diagonal translation by $a_0 = 1 \times w_0 \in S_{2n}$, where w_0 is the longest element in the second block: it identifies $\tau X_u \times X_v$ with $w_0^{(2n)} X_u \times X_v$. The codimension is therefore

$$\mathrm{codim}_X(w_0^{(2n)} X_u) + \ell(v) = \ell(u) + \ell(v).$$

By the incidence formula of Mihalcea [12, Section 3.1], this gives

$$\mathrm{ev}_1^*[\tau X_u]^T \ \mathrm{ev}_2^*[X_v]^T \cap [M_d]^T = [I^d_{u,v}]_T. \tag{2.3}$$

The cycle $\Gamma^d_{u,v}$ is effective and $B^-(\tau)$-invariant. To see invariance, write $N^-(\tau) = N^- \cap \tau N^- \tau^{-1}$ and $B^-(\tau) = TN^-(\tau)$. The varieties X_v and τX_u are stable under the corresponding actions, so their product, its inverse image under the equivariant map F, and its proper push-forward under ev_3 are $B^-(\tau)$-invariant. Effectivity follows from the reducedness of $I^d_{u,v}$.

There is a unique expansion

$$[\Gamma^d_{u,v}]_T = \sum_{r \in S_{2n}} p^{r,d}_{u,v}(\mathbf{a})[X_r]_T \quad \text{in } H^*_T(X).$$

By Gao and Xiong's Graham–positivity theorem for $B^-(\tau)$-invariant effective cycles [6, Theorem 2.3 and Corollary 2.4],

$$p^{r,d}_{u,v}(\mathbf{a}) \in \mathbb{Z}_{\geq 0}[t_i - y_j : 1 \leq i,j \leq n].$$

Finally, by equivariant Poincaré duality, (2.3), and the projection formula,

$$(\mathrm{ev}_3)^T_* \left(\mathrm{ev}_1^*[\tau X_u]^T \, \mathrm{ev}_2^*[X_v]^T\right) = [\Gamma^d_{u,v}]_T,$$

and hence the degree-d summand of (2.2) is

$$(\pi_X)^T_*\left([\Gamma^d_{u,v}]_T [Y_w]^T\right) = \sum_{r \in S_{2n}} p^{r,d}_{u,v}(\mathbf{a})(\pi_X)^T_*\left([X_r]^T [Y_w]^T\right) = p^{w,d}_{u,v}(\mathbf{a}).$$

Thus every coefficient $C^{w,d}_{u,v}(y,t)$ is in $\mathbb{Z}_{\geq 0}[t_i - y_j]$. Replacing (u,v,y,t) by (v,u,z,y) proves the asserted positivity of $c^{w,d}_{u,v}(y;z)$. □

Example 2.4. We compute the product $\widetilde{\mathfrak{S}}_{15423}(x;y)\,\widetilde{\mathfrak{S}}_{1432}(x;z)$:

$$\begin{aligned}
&q_2q_3\,(y_1+y_4-z_1-z_2)\,\widetilde{\mathfrak{S}}_{124635}\,(x;y) + q_2q_3\,(y_1+y_4-z_1-z_2)\,\widetilde{\mathfrak{S}}_{13452}\,(x;y)\\
&+ q_2q_3\,(y_1+y_4-z_1-z_2)\,\widetilde{\mathfrak{S}}_{14253}\,(x;y)\\
&+ q_2q_3\,\widetilde{\mathfrak{S}}_{125634}\,(x;y) + q_2q_3\,\widetilde{\mathfrak{S}}_{126435}\,(x;y) + q_2q_3\,\widetilde{\mathfrak{S}}_{13542}\,(x;y) + q_2q_3\,\widetilde{\mathfrak{S}}_{214635}\,(x;y)\\
&+ q_2q_3\,\widetilde{\mathfrak{S}}_{23451}\,(x;y) + q_2q_3\,\widetilde{\mathfrak{S}}_{24153}\,(x;y) + q_2q_3\,\widetilde{\mathfrak{S}}_{31452}\,(x;y) + q_2q_3\,\widetilde{\mathfrak{S}}_{41253}\,(x;y)\\
&+ q_2\,(y_1+y_4-z_1-z_2)\,\widetilde{\mathfrak{S}}_{146235}\,(x;y) + q_2\,(y_4+y_5-z_1-z_2)\,\widetilde{\mathfrak{S}}_{24513}\,(x;y)\\
&+ q_2\,\widetilde{\mathfrak{S}}_{246135}\,(x;y) + q_3\,\widetilde{\mathfrak{S}}_{1724356}\,(x;y) + q_3\,\widetilde{\mathfrak{S}}_{261435}\,(x;y) + q_3\,\widetilde{\mathfrak{S}}_{35142}\,(x;y)\\
&+ \left(q_2q_3 + q_2\left((y_1-z_2)\,(y_4+y_5-z_1-z_2) + (y_4-z_1)\,(y_5-z_1)\right)\right)\widetilde{\mathfrak{S}}_{14523}\,(x;y)\\
&+ \big((y_4-z_1)\,(y_5-z_3) + (y_4-z_1)\,(y_1+y_2-z_1-z_2)\\
&\quad + (y_5-z_2)\,(y_5-z_3) + (y_5-z_2)\,(y_1+y_2-z_1-z_2)\big)\,\widetilde{\mathfrak{S}}_{25413}\,(x;y)\\
&+ \big((y_1-z_2)\,(y_1-z_3)\,(y_4-z_2) + (y_1-z_2)\,(y_1-z_3)\,(y_5-z_1) + (y_1-z_2)\,(y_5-z_1)\,(y_5-z_2)\\
&\quad + (y_1-z_3)\,(y_4-z_1)\,(y_5-z_1) + (y_4-z_1)\,(y_5-z_1)\,(y_5-z_2)\big)\,\widetilde{\mathfrak{S}}_{15423}\,(x;y)\\
&+ \left(q_2 + (y_1-z_2)\,(y_1-z_3) + (y_1-z_3)\,(y_5-z_1) + (y_5-z_1)\,(y_5-z_2)\right)\widetilde{\mathfrak{S}}_{156234}\,(x;y)\\
&+ \left(q_3 + (y_1-z_1)\,(y_4-z_1) + (y_5-z_2)\,(y_1+y_4-z_1-z_2)\right)\widetilde{\mathfrak{S}}_{45123}\,(x;y)
\end{aligned}$$

$$
\begin{aligned}
&+ (q_2q_3 + q_3((y_1 - z_1)(y_1 + y_5 - z_2 - z_3) + (y_5 - z_2)(y_5 - z_3)))\,\widetilde{\mathfrak{S}}_{15243}(x;y)\\
&+ q_3(y_1 + y_5 + y_6 - z_1 - z_2 - z_3)\,\widetilde{\mathfrak{S}}_{162435}(x;y) + q_3(y_1 + y_2 + y_5 - z_1 - z_2 - z_3)\,\widetilde{\mathfrak{S}}_{25143}(x;y)\\
&+ q_2q_3(y_1 + y_2 + y_5 - z_1 - z_2 - z_3)\,\widetilde{\mathfrak{S}}_{12543}(x;y) + q_2q_3(y_1 + y_2 + y_5 - z_1 - z_2 - z_3)\,\widetilde{\mathfrak{S}}_{21453}(x;y)\\
&+ q_2q_3((y_1 - z_2)(y_1 - z_3) + (y_1 - z_2)(y_2 - z_2) + (y_1 - z_2)(y_4 - z_1)\\
&\quad + (y_1 - z_2)(y_5 - z_1) + (y_2 - z_1)(y_4 - z_1) + (y_4 - z_1)(y_5 - z_3))\,\widetilde{\mathfrak{S}}_{12453}(x;y)\\
&+ (y_1 + y_4 - z_1 - z_2)\,\widetilde{\mathfrak{S}}_{1742356}(x;y) + (y_1 + y_4 - z_1 - z_2)\,\widetilde{\mathfrak{S}}_{461235}(x;y)\\
&+ (y_4 + y_5 - z_1 - z_2)\,\widetilde{\mathfrak{S}}_{35412}(x;y) + (y_4 + y_5 - z_1 - z_2)\,\widetilde{\mathfrak{S}}_{45213}(x;y)\\
&+ ((y_1 - z_2)(y_1 - z_3) + (y_1 - z_2)(y_5 + y_6 - z_1 - z_2)\\
&\quad + (y_1 - z_3)(y_4 - z_1) + (y_4 - z_1)(y_5 + y_6 - z_1 - z_2))\,\widetilde{\mathfrak{S}}_{164235}(x;y)\\
&+ (y_1 + y_2 + y_5 - z_1 - z_2 - z_3)\,\widetilde{\mathfrak{S}}_{256134}(x;y) + (y_1 + y_5 + y_6 - z_1 - z_2 - z_3)\,\widetilde{\mathfrak{S}}_{165234}(x;y)\\
&+ (y_1 + y_2 + y_4 + y_5 + y_6 - 2z_1 - 2z_2 - z_3)\,\widetilde{\mathfrak{S}}_{264135}(x;y)\\
&+ \widetilde{\mathfrak{S}}_{1752346}(x;y) + \widetilde{\mathfrak{S}}_{265134}(x;y) + \widetilde{\mathfrak{S}}_{2741356}(x;y)\\
&+ \widetilde{\mathfrak{S}}_{356124}(x;y) + \widetilde{\mathfrak{S}}_{364125}(x;y) + \widetilde{\mathfrak{S}}_{462135}(x;y)\\
&+ \widetilde{\mathfrak{S}}_{561234}(x;y)
\end{aligned}
$$

By inspection, we can see that the coefficients exhibit triple Schubert positivity.

Remark 2.5. The Pieri formula of Theorem 3.4 below is an explicit instance of Theorem 2.3: for $v = s_1 s_2 \cdots s_k$ each coefficient $c_{u,v}^{w,d}(y;z)$ is a product of differences $y_a - z_1$. The parabolic reduction of Proposition 2.10 shows that the structure constants of the parabolic quantum double Schubert polynomials of the next subsection are also triple Schubert positive (Corollary 2.12).

2.2. **The parabolic case.** Unlike ordinary cohomology or equivariant cohomology, quantum cohomology and equivariant quantum cohomology do not demonstrate simple functoriality, in the sense that a projection of the complete flag variety G/B to a partial flag variety G/P does not induce a homomorphism of rings $QH_T^*(G/P) \to QH_T^*(G/B)$, and indeed the quantum cohomology of G/P is not isomorphic to the subring of the quantum cohomology of G/B generated by the Schubert classes indexed by W^P. This is reflected in the fact that there are parabolic quantum double Schubert polynomials defined in a different manner that represent Schubert classes in $QH_T^*(G/P)$ [10] and are not in general equal to the quantum double Schubert polynomials above.

Definition 2.6 (Parabolic quantum double Schubert polynomials)**.** Following Lam and Shimozono [10], we define parabolic quantum double Schubert polynomials as follows. Fix an integer $n > 0$. A parabolic subgroup W_P of S_n, which has Coxeter generators $s_1, \ldots, s_{n-1}$, is generated by a subset $P \subseteq \{s_1, \ldots, s_{n-1}\}$. In terms of the subgroup's properties, it is most natural to identify P by the reflections that it *excludes*. If P is generated by $S \setminus \{s_{i_1}, \ldots, s_{i_k}\}$ in increasing order, we define integers $n_1, \ldots, n_{k+1}$ with

$$n_1 + \cdots + n_{k+1} = n$$

by $n_1 = i_1$ and $n_p = i_p - i_{p-1}$ for $2 \le p \le k+1$. Then N_j for $j \ge 1$ is defined by

$$N_j = n_1 + \cdots + n_j$$

for $0 \le j \le k+1$.

For such a setup, define a matrix D_j^P to be the $n \times n$ matrix with $x_1, \ldots, x_n$ on the diagonal, -1 on the superdiagonal, and $(-1)^{n_{j+1}+1} q_j$ at $(N_{j+1}, N_{j-1} + 1)$ for $1 \le j \le k$. Notice that there are precisely k q variables, and we set

$$\deg q_j = n_j + n_{j+1}$$

Define

$$\Delta_j^P(t;x) = \det(D_j^P - t\mathrm{Id})$$

Then we define the parabolic quantum double Schubert polynomials $\widetilde{\mathfrak{S}}^P_w(x;y)$ for $w \in W^P$ by

$$\widetilde{\mathfrak{S}}^P_w(x;y) = (-1)^{\ell(w(w_0^P)^{-1})}\partial^{(y)}_{w(w_0^P)^{-1}} \prod_{j=1}^{k} \Delta^P_j(-y_{n-N_{j+1}+1}, \ldots, -y_{n-N_j}; x)$$

where w_0^P is the longest element of W^P.

Taken as defined, when we vary n these polynomials are not stable under inclusion and do not form a basis of $\mathbb{Z}[q][x_1, \ldots, x_n; y]$ as a module over $\mathbb{Z}[q][y]$. To obtain stability and a basis, instead we fix n and choose $N > n$ as large as we would like, maintaining $N_{k+1} = n$ and setting $n_{k+2} = N - n$. From a cohomological perspective, this introduces "extraneous" polynomials that are indexed by elements not in W^P. The full indexing set is $W^{\widehat{P}} \subset S_\infty$, where

$$\widehat{P} = P \cup \{s_{n+1}, s_{n+2}, \ldots\}$$

Example 2.7. As equivariant cohomology rings of finite dimensional projective varieties over $\mathbb{C}$ are finitely generated modules over the coefficient ring, it is fully expected that products of double Schubert polynomials may include terms indexed by permutations that do not lie within the indexing set of the basis of Schubert classes. For example, consider the following non-quantum double Schubert polynomial product, with $P = \{s_1, s_3\}$ and basis elements in S_4:

$$\begin{aligned}\mathfrak{S}_{2413}(x;y)\,\mathfrak{S}_{3412}(x;y) = &-(\alpha_1+\alpha_2)(\alpha_1+\alpha_2+\alpha_3)(\alpha_2+\alpha_3)\ \mathfrak{S}_{3412}(x;y)\\ &+(\alpha_1+\alpha_2)(\alpha_1+2\alpha_2+2\alpha_3+\alpha_4)\ \mathfrak{S}_{35124}(x;y)\\ &-(\alpha_1+\alpha_2)\ \mathfrak{S}_{361245}(x;y)\\ &-(\alpha_1+2\alpha_2+2\alpha_3+\alpha_4)\ \mathfrak{S}_{45123}(x;y)\\ &+\mathfrak{S}_{461235}(x;y).\end{aligned}$$

Not only does the product $\widetilde{\mathfrak{S}}^P_{2413}(x;y)\,\widetilde{\mathfrak{S}}^P_{3412}(x;y)$ include extra terms indexed by permutations in W^P, there are also terms with nonzero coefficients that are in $W^{\widehat{P}} \setminus W^P$:

$$\begin{aligned}q_1(\alpha_1\alpha_2+\alpha_1\alpha_3+\alpha_2\alpha_3+\alpha_2^2)\ \widetilde{\mathfrak{S}}^P_{132}(x;y) + q_1(\alpha_1+\alpha_2)\ &\widetilde{\mathfrak{S}}^P_{231}(x;y)\\ &+ q_1(\alpha_2+\alpha_3)\ \widetilde{\mathfrak{S}}^P_{1423}(x;y)\\ &+ q_1(\alpha_1+\alpha_2)\ \widetilde{\mathfrak{S}}^{\widehat{P}}_{13254}(x;y)\\ &+ q_1\ \widetilde{\mathfrak{S}}^{\widehat{P}}_{14253}(x;y)\\ &+ q_1\ \widetilde{\mathfrak{S}}^P_{15234}(x;y)\\ &+ q_1\ \widetilde{\mathfrak{S}}^P_{2413}(x;y)\end{aligned}$$

Definition 2.8 (q-vectors and Peterson-Woodward lifts)**.** Define $Q^\vee$, for legacy purposes, to be the set of weak compositions. For an integer i and $d \in Q^\vee$, define

$$\langle d\rangle_i = 2d_i - d_{i-1} - d_{i+1}$$

The double indexed version will be denoted by

$$\langle d\rangle_{ij} = \sum_{k=i}^{j-1} \langle d\rangle_k$$

For a parabolic subgroup W_P, we define $Q^\vee_P$ to be the set of weak compositions d such that $\langle d\rangle_i = 0$ for all $s_i \notin P$. The q-vectors in W^P are indexed by $Q^\vee/Q^\vee_P$.

Let $d_P \in Q^\vee/Q^\vee_P$. A *Peterson-Woodward lift* of d_P is a $d \in Q^\vee$ such that $d_P \in d + Q^\vee_P$ and $\langle d\rangle_{ij} \in \{0, -1\}$ for all i, j such that $s_i, \ldots, s_{j-1} \in P$.

Method 2.9 (Checking the Peterson-Woodward lift property)**.** While backing out a Peterson-Woodward lift is difficult, going in the other direction is not. Suppose $d \in Q^\vee$ and P is parabolic. If there is any $i \in P$ such that $\langle d\rangle_i \notin \{0, -1\}$, then d is not a Peterson-Woodward lift. If this condition does hold, let $I_1, \ldots, I_m$ be the maximal intervals of consecutive indices i with $s_i \in P$. Since $\langle d\rangle_{ij} = \sum_{l=i}^{j-1} \langle d\rangle_l$, the condition $\langle d\rangle_{ij} \in \{0, -1\}$

for all $i < j$ with $s_i, \ldots, s_{j-1} \in P$ holds if and only if each interval I_r contains at most one index l with $\langle d \rangle_l = -1$. If and only if this is the case, d is the Peterson-Woodward lift of a unique $d_P \in Q^\vee/Q_P^\vee$, namely its image, which we denote by $\pi_P(d)$. In that case we let

$$z_P(d) = \{\, i \mid s_i \in P,\ \langle d \rangle_i = 0 \,\}$$

and write $w_{z_P(d)}$ for the longest element of the parabolic subgroup generated by $\{s_i \mid i \in z_P(d)\}$, and w_P for the longest element of W_P.

This algorithm is polynomial in the length of d, which in almost all currently feasible cases is quite small. In contrast, a naive search for d with d_P as the input has an exponential search space, though a more refined algorithm may exist.

The parabolic triple quantum Schubert structure constants $c_{u,v}^{w,d_P}(y;z)$ are defined by

$$\widetilde{\mathfrak{S}}_u^P(x;y)\,\widetilde{\mathfrak{S}}_v^P(x;z) = \sum_{w \in W^{\widehat{P}},\, d_P} q^{d_P}\, c_{u,v}^{w,d_P}(y;z)\ \widetilde{\mathfrak{S}}_w^P(x;y).$$

We prove the following result in Section 8, as a consequence of the Peterson–Woodward comparison formula of Huang and Li [7] and Lemma 2.1.

Proposition 2.10. *Let W_P be a parabolic subgroup of S_∞ with $P = \widehat{P}$ and let $u, v, w \in W^P$. Then for any $d_P \in Q^\vee/Q_P^\vee$ with Peterson–Woodward lift d, we have that*

$$c_{u,v}^{w,d_P}(y;z) = c_{u,v}^{ww_Pw_{z_P(d)},d}(y;z).$$

Thus, though we lack a simple functoriality property, we can reduce the computation of parabolic triple quantum Schubert structure constants completely to the complete flag variety case with an efficient algorithm. We obtain a nontrivial corollary from this.

Example 2.11. The product in the example above, with $P = \widehat{P}$ satisfying $P \cap \{s_1, s_2, s_3, s_4, s_5\} = \{s_1, s_4\}$, allows us to compute the triple structure constants for the corresponding parabolic quantum double Schubert polynomials:

$$\begin{aligned}
&q_1q_2\,(y_1+y_2+y_5-z_1-z_2-z_3)\,\widetilde{\mathfrak{S}}_{1243}^P(x;y)\\
&+ q_1q_2\,\widetilde{\mathfrak{S}}_{124365}^P(x;y) + q_1q_2\,\widetilde{\mathfrak{S}}_{1342}^P(x;y) + q_1q_2\,\widetilde{\mathfrak{S}}_{1423}^P(x;y)\\
&+ q_2\,((y_1-z_2)(y_1-z_3)+(y_1-z_3)(y_5-z_1)+(y_5-z_1)(y_5-z_2))\,\widetilde{\mathfrak{S}}_{15234}^P(x;y)\\
&+ q_2\,(y_1+y_5+y_6-z_1-z_2-z_3)\,\widetilde{\mathfrak{S}}_{162345}^P(x;y) + q_2\,(y_1+y_2+y_5-z_1-z_2-z_3)\,\widetilde{\mathfrak{S}}_{25134}^P(x;y)\\
&+ ((y_1-z_2)(y_1-z_3)(y_4-z_2)+(y_1-z_2)(y_1-z_3)(y_5-z_1)+(y_1-z_2)(y_5-z_1)(y_5-z_2)\\
&\quad +(y_1-z_3)(y_4-z_1)(y_5-z_1)+(y_4-z_1)(y_5-z_1)(y_5-z_2))\,\widetilde{\mathfrak{S}}_{15423}^P(x;y)\\
&+ q_2\,\widetilde{\mathfrak{S}}_{1723456}^P(x;y) + q_2\,\widetilde{\mathfrak{S}}_{261345}^P(x;y) + q_2\,\widetilde{\mathfrak{S}}_{35124}^P(x;y)\\
&+ ((y_1-z_2)(y_1-z_3)+(y_1-z_3)(y_5-z_1)+(y_5-z_1)(y_5-z_2))\,\widetilde{\mathfrak{S}}_{156234}^P(x;y)\\
&+ ((y_1-z_2)(y_1-z_3)+(y_1-z_2)(y_5+y_6-z_1-z_2)\\
&\quad +(y_1-z_3)(y_4-z_1)+(y_4-z_1)(y_5+y_6-z_1-z_2))\,\widetilde{\mathfrak{S}}_{164235}^P(x;y)\\
&+ ((y_4-z_1)(y_5-z_3)+(y_4-z_1)(y_1+y_2-z_1-z_2)\\
&\quad +(y_5-z_2)(y_5-z_3)+(y_5-z_2)(y_1+y_2-z_1-z_2))\,\widetilde{\mathfrak{S}}_{25413}^P(x;y)\\
&+ ((y_1-z_1)(y_4-z_1)+(y_5-z_2)(y_1+y_4-z_1-z_2))\,\widetilde{\mathfrak{S}}_{45123}^P(x;y)\\
&+ (y_1+y_5+y_6-z_1-z_2-z_3)\,\widetilde{\mathfrak{S}}_{165234}^P(x;y) + (y_1+y_2+y_5-z_1-z_2-z_3)\,\widetilde{\mathfrak{S}}_{256134}^P(x;y)\\
&+ (y_1+y_4-z_1-z_2)\,\widetilde{\mathfrak{S}}_{1742356}^P(x;y) + (y_1+y_4-z_1-z_2)\,\widetilde{\mathfrak{S}}_{461235}^P(x;y)\\
&+ (y_1+y_2+y_4+y_5+y_6-2z_1-2z_2-z_3)\,\widetilde{\mathfrak{S}}_{264135}^P(x;y)
\end{aligned}$$

$$+ (y_4 + y_5 - z_1 - z_2)\,\widetilde{\mathfrak{S}}^P_{35412}(x;y) + (y_4 + y_5 - z_1 - z_2)\,\widetilde{\mathfrak{S}}^P_{45213}(x;y)$$
$$+ \widetilde{\mathfrak{S}}^P_{1752346}(x;y) + \widetilde{\mathfrak{S}}^P_{265134}(x;y) + \widetilde{\mathfrak{S}}^P_{2741356}(x;y)$$
$$+ \widetilde{\mathfrak{S}}^P_{356124}(x;y) + \widetilde{\mathfrak{S}}^P_{364125}(x;y) + \widetilde{\mathfrak{S}}^P_{462135}(x;y)$$
$$+ \widetilde{\mathfrak{S}}^P_{561234}(x;y)$$

For an illustration, consider $d_P = (-,0,1,-,0)$ for the coefficient of 162345. $q^{d_P} = q_2$. The Peterson-Woodward lift of d_P is $d = (0,0,1,0,0)$, and $w_P = 21354$. $z_P(d) = \{1\}$, so $w_P w_{z_P(d)} = 12354$. Thus, $w w_P w_{z_P(d)} = 162435$, and the q_2-component of the coefficient of $\widetilde{\mathfrak{S}}^P_{162345}(x;y)$ is $(y_1 + y_5 + y_6 - z_1 - z_2 - z_3)$, which is the same as the q_3-component (q^d) of the coefficient of $\widetilde{\mathfrak{S}}_{162435}(x;y)$ in the complete flag variety case (Example 2.4).

Corollary 2.12 (Triple quantum Schubert positivity for the parabolic case)**.** *Let W_P be a parabolic subgroup with $P = \widehat{P}$ and let $u, v, w \in W^P$. Then for every d_P the parabolic structure constant $c^{w,d_P}_{u,v}(y;z)$ exhibits triple Schubert positivity. In particular this holds when u, v, w are k-Grassmannian and P is the corresponding maximal parabolic subgroup ($n_1 = k$, $n_2 = n - k$).*

Proof. By Proposition 2.10, $c^{w,d_P}_{u,v}(y;z)$ equals a structure constant $c^{w',d}_{u,v}(y;z)$ of the complete flag variety, which lies in $\mathbb{Z}_{\geq 0}[y_i - z_j]$ by Theorem 2.3. □

3. Pieri

Definition 3.1 ($\overset{k}{\rightsquigarrow}$, $\mathrm{fix}_k(u,w)$, $q_k(u,w)$, $E_{p,k}(y;z)$, $\mathbf{E}(y_A;z)$)**.** We define a relation $\overset{k}{\rightsquigarrow}$ on pairs of permutations, introduced in its non-quantum form by Sottile [19], which was used originally to indicate when the coefficient of a Schubert polynomial is nonzero in a suitable application of the Pieri formula. In the generalization to double Schubert polynomials in [16], the same relation applies to all nonzero coefficients in the expansion of the generalized Pieri formula. The relation plays the same role in our case.

Given $u, u' \in S_\infty$ and a positive integer k, we declare that $u \overset{k}{\rightsquigarrow} w$ if there exists a p with $0 \leq p \leq k$ such that there are transpositions $t_{a_1 b_1}, \ldots, t_{a_p b_p}$ satisfying

(1) $a_i \leq k < b_i$ for all i
(2) $a_i \neq a_j$ whenever $i \neq j$
(3) $b_i \geq b_j$ whenever $i < j$
(4) We have either
$$\ell(u t_{a_1 b_1} \cdots t_{a_i b_i}) = \ell(u t_{a_1 b_1} \cdots t_{a_{i-1} b_{i-1}}) + 1$$
or
$$\ell(u t_{a_1 b_1} \cdots t_{a_i b_i}) = \ell(u t_{a_1 b_1} \cdots t_{a_{i-1} b_{i-1}}) - 2(b_i - a_i) + 1$$
for all i
(5) $w = u t_{a_1 b_1} \cdots t_{a_p b_p}$

In contrast to the non-quantum case, $u \overset{k}{\rightsquigarrow} w$ does not imply that $\ell(u) \leq \ell(w)$, as the final condition can cause the length of the permutation to drop. For intuition, the condition $\ell(u t_{ab}) = \ell(u) - 2(b-a) + 1$ means that if we write the reflection t_{ab} as
$$s_b s_{b-1} \cdots s_{a+1} s_a s_{a+1} \cdots s_{b-1} s_b$$
then each simple reflection in this word cancels a corresponding simple reflection in a reduced word for u, as opposite to the length-increase case where there is a word for which precisely one simple reflection is inserted, regardless of the length of the reflection as a permutation.

If $u \overset{k}{\rightsquigarrow} w$, define
$$\mathrm{fix}_k(u,w) = \{u(i) \mid i \leq k \text{ and } u(i) = w(i)\}$$
and
$$n_k(u,w) = k - |\mathrm{fix}_k(u,w)|.$$

Also, we define $q_k(u,w)$ recursively. If $u = w$, define $q_k(u,w) = 1$. If we have defined $q_k(u,w')$ with $w' = t_{a_1b_1}\cdots t_{a_{p-1}b_{p-1}}$, suppose $\ell(w't_{a_pb_p}) = \ell(w') + 1$. Then define

$$q_k(u,w) = q_k(u,w')$$

If instead $\ell(w't_{a_pb_p}) = \ell(w') - 2(b_p - a_p) + 1$, define

$$q_k(u,w) = q_k(u,w')q_{a_pb_p}$$

Suppose y and z are sets of variables and $0 \leq p \leq k$ are integers. We define $E_{p,k}(y;z)$ to be the factorial elementary symmetric polynomial in y and z. For specificity, this is the double Schubert polynomial $\mathfrak{S}_{c[p,k]}(y;z)$ where

$$c[p,k] = s_{k-p+1}\cdots s_{k-1}s_k$$

If $p = 0$, the polynomial is identically 1. If $p = k$, $E_{k,k}(y;z)$ can be simply written as

$$E_{k,k}(y;z) = \prod_{i=1}^{k}(y_i - z_1)$$

An explicit formula can be written for general p without too much trouble as well, using e.g. [14]:

$$E_{p,k}(y;z) = \sum_{1\leq i_1<\cdots<i_p\leq k}\prod_{j=1}^{p}(y_{i_j} - z_{i_j-j+1})$$

For a set $A = \{i_1,\ldots,i_{|A|}\}$, we define

$$\mathbf{E}(y_A;z) = \prod_{i\in A}(y_i - z_1) = E_{|A|,|A|}(y_{i_1},\ldots,y_{i_{|A|}};z),$$

the specialization of the factorial elementary polynomial to which the coefficients of the Pieri formula reduce. It will be used to express those coefficients.

Definition 3.2 (Quantum factorial elementary polynomials $\widetilde{E}_{p,k}(x;z)$)**.** For $0 \leq p \leq k$ we define

$$\widetilde{E}_{p,k}(x;z) = \widetilde{\mathfrak{S}}_{c[p,k]}(x;z),$$

the quantum double Schubert polynomial of the Grassmannian permutation $c[p,k] = s_{k-p+1}\cdots s_k$, whose diagram is a single column of height p with descent at k. At $q = 0$ this is $E_{p,k}(x;z)$. We also set $\widetilde{E}_{0,k} = 1$ and $\widetilde{E}_{p,k} = 0$ for $p < 0$.

Lemma 3.3. *For $k \geq 1$,*

$$\widetilde{E}_{k,k}(x;z) = \Delta_k(-z_1 \mid x_1,\ldots,x_k) = \sum_{j=0}^{k}(-z_1)^{k-j}e_j^{(q)}(x_1,\ldots,x_k).$$

Proof. The permutation $c[k,k] = s_1s_2\cdots s_k$ has a unique reduced word, so its reduced factorizations are $v_1 = s_1\cdots s_j$, $v_2 = s_{j+1}\cdots s_k$ for $0 \leq j \leq k$. By Lemma 2.1 with $y = 0$, $\widetilde{E}_{k,k}(x;z) = \sum_j \mathfrak{S}_{s_1\cdots s_j}(0;z)\ \widetilde{\mathfrak{S}}_{s_{j+1}\cdots s_k}(x)$. Here $s_1\cdots s_j$ is dominant with code (1^j), so $\mathfrak{S}_{s_1\cdots s_j}(x;z) = \prod_{i\leq j}(x_i - z_1)$ and $\mathfrak{S}_{s_1\cdots s_j}(0;z) = (-z_1)^j$; and $s_{j+1}\cdots s_k$ is the Grassmannian permutation with descent at k and a single column of height $k-j$, whose Schubert polynomial $e_{k-j}(x_1,\ldots,x_k)$ is a standard elementary monomial, so that its quantum Schubert polynomial is $e^{(q)}_{k-j}(x_1,\ldots,x_k)$ [5]. Summing gives the second expression, which equals $\Delta_k(-z_1 \mid x_1,\ldots,x_k)$ by the definition of the quantum elementary polynomials. □

Theorem 3.4 (The Pieri formula)**.** *Suppose $u \in S_\infty$ and $k \geq 1$. Then*

$$\widetilde{\mathfrak{S}}_u(x;y)\widetilde{E}_{k,k}(x;z) = \sum_{u\overset{k}{\rightsquigarrow}w} q_k(u,w)\mathbf{E}(y_{\mathrm{fix}_k(u,w)};z)\,\widetilde{\mathfrak{S}}_w(x;y).$$

The proof is given in Section 6. Note that the right-hand side is manifestly triple Schubert positive, in accordance with Theorem 2.3, and that at $q = 0$ it reduces to the case $p = k$ of the Pieri formula of [16, Theorem 7.1].

4. Algorithms for multiplying (quantum) double Schubert polynomials

Definition 4.1 ($\mathfrak{c}$, (strict) dominant permutations, θ, μ, $\widetilde{\theta}$, $\widetilde{\mu}$, medium theta $\overline{\theta}$, $\overline{\mu}$)**.** We say that a permutation μ is *dominant* if $\mathfrak{c}_i(\mu) \geq \mathfrak{c}_{i+1}(\mu)$ for all i. A permutation μ is said to be *strict dominant* if μ is dominant and whenever $\mathfrak{c}_i(\mu) \neq 0$ we have $\mathfrak{c}_i(\mu) > \mathfrak{c}_{i+1}(\mu)$. For a permutation $v \in S_\infty$, we define a dominant permutation $\mu(v)$ recursively as follows. If v is already dominant, define $\mu(v) = v$. Otherwise, let i be the maximal index such that $\mathfrak{c}_i(v) < \mathfrak{c}_{i+1}(v)$, and define

$$\mu(v) = \mu(vs_i)$$

Define $\theta_i(v)$ to be $\mathfrak{c}_i(\mu(v))$.

Similarly, for a permutation $v \in S_\infty$, we define a strict dominant permutation $\widetilde{\mu}(v)$ recursively as follows. If v is already strict dominant, define $\widetilde{\mu}(v) = v$. Otherwise, let i be the maximal index such that $\mathfrak{c}_{i+1}(v) \neq 0$ and $\mathfrak{c}_i(v) \leq \mathfrak{c}_{i+1}(v)$, then define

$$\widetilde{\mu}(v) = \widetilde{\mu}(vs_i)$$

Define $\widetilde{\theta}_i(v)$ to be $\mathfrak{c}_i(\widetilde{\mu}(v))$.

Between these two lies the *medium theta* $\overline{\theta}(v)$, the partition actually iterated by the quantum multiplication algorithm. Like $\theta(v)$ it is a weakly decreasing partition obtained from $\mathfrak{c}(v)$ by incrementing and reordering, but it stops short of separating every pair of equal parts: a pair of equal consecutive nonzero parts $\overline{\theta}_i(v) = \overline{\theta}_{i+1}(v)$ is left in place exactly when it is preceded by a strict drop of at least two, that is $\overline{\theta}_{i-1}(v) \geq \overline{\theta}_i(v) + 2$ (or $i = 1$), and every other ascent or equality is removed by incrementing as for θ. Thus $\overline{\theta}(v)$ permits some, but not all, repeated parts, and lies between $\theta(v)$ (arbitrary repeats) and $\widetilde{\theta}(v)$ (no repeats). We write $\overline{\mu}(v)$ for the permutation with $\mathfrak{c}(\overline{\mu}(v)) = \overline{\theta}(v)$ and set $|\overline{\theta}(v)| := \sum_i \overline{\theta}_i(v) = \ell(\overline{\mu}(v))$. The number of nonzero parts of $\overline{\theta}(v)$ equals the number of nonzero parts of $\mathfrak{c}(v)$.

Definition 4.2 ($\overset{i}{\Rightarrow}$, $\sigma_i(u,v)$, $D_i(u,v)$)**.** Let $u, v \in S_\infty$ and $i \geq 1$ be an integer. We define $u \overset{i}{\Rightarrow} v$ if there exist distinct integers $b_1, b_2, \ldots, b_k$ such that $b_j \neq i$ for all j, $\ell(ut_{i,b_1} \cdots t_{i,b_j}) = \ell(u) + j$ for all j, and

$$v = ut_{i,b_1} \cdots t_{i,b_k}$$

We define

$$\sigma_i(u,v) = \#\{j \mid b_j < i\}$$

and we define

$$D_i(u,v) = \{u(i)\} \cup \{u(j) \mid u(j) \neq v(j)\}$$

The following two formulas express a (quantum) double Schubert polynomial as a signed sum of products of (quantum) factorial elementary polynomials in nested initial alphabets; they are the v-side input of the multiplication algorithm. The first is proved in [17]. The second is its quantum analogue and is proved in the same way, the only new ingredient being the factorization of $\widetilde{\mathfrak{S}}_\mu(x;z)$ for a strict dominant permutation μ: if the column heights $\lambda'_1 > \lambda'_2 > \cdots$ of the diagram of μ are pairwise distinct, then $x^{\mathfrak{c}(\mu)} = \prod_j e_{\lambda'_j}(x_1, \ldots, x_{\lambda'_j})$ is a single standard elementary monomial, so that the Fomin–Gelfand–Postnikov quantization gives $\widetilde{\mathfrak{S}}_\mu(x;z) = \prod_j \widetilde{E}_{\lambda'_j,\lambda'_j}(x;z_j)$; this is the reason μ is replaced by $\widetilde{\mu}$ in the quantum case, since for repeated column heights the quantum polynomial is *not* a product (see Section 7).

Theorem 4.3 (Formula for ordinary double Schubert polynomials, [17])**.** *Suppose $v \in S_\infty$. Then*

$$\mathfrak{S}_v(x;z) = \sum_{(v_0,\ldots,v_k)} \prod_{i=1}^{k} (-1)^{\sigma_i(v_{i-1},v_i)} E_{\theta_i(v)-\ell(v_{i-1},v_i),\theta_i(v)}(x; z_{D_i(v_{i-1},v_i)})$$

Where $v_{i-1} \overset{i}{\Rightarrow} v_i$ for all i, $v_0 = 1$, and $v_k = v\mu(v^{-1})$.

Theorem 4.4 (Formula for quantum double Schubert polynomials)**.** *Suppose $v \in S_\infty$. Then*

$$\widetilde{\mathfrak{S}}_v(x;z) = \sum_{(v_0,\ldots,v_k)} \prod_{i=1}^{k} (-1)^{\sigma_i(v_{i-1},v_i)} E^q_{\widetilde{\theta}_i(v)-\ell(v_{i-1},v_i),\widetilde{\theta}_i(v)}(x; z_{D_i(v_{i-1},v_i)})$$

Where $v_{i-1} \overset{i}{\Rightarrow} v_i$ for all i, $v_0 = 1$, and $v_k = v\widetilde{\mu}(v^{-1})$.

Coupled with the Pieri formulas, these formulas give an algorithm for computing products of (quantum) double Schubert polynomials, which is the one implemented in [18]. We describe its classical form by building it from two ingredients: the elementary Pieri step that generates the u-side covers, and the signed path expansion on the v-side.

Definition 4.5 (k-strip cover, admissible k-chain, the (p,k)-Pieri problem)**.** Let $w \in S_n$, let $1 \le k < n$, and let $\beta \in \{1,\dots,n\} \cup \{\infty\}$. A *$k$-strip cover of* (w,β) is a transposition t_{ab} of positions a, b with $1 \le a \le k < b \le n$ such that

(1) $w \lessdot wt_{ab}$ in Bruhat order; that is, $w(a) < w(b)$ and no c with $a < c < b$ satisfies $w(a) < w(c) < w(b)$; and
(2) $w(b) < \beta$.

Its *output* is the pair $(wt_{ab},\, w(b))$. An *admissible k-chain of length p* from u is a sequence $u = w_0, w_1, \dots, w_p$ carrying bounds $\infty = \beta_0 > \beta_1 > \cdots > \beta_p$ in which (w_t, β_t) is the output of a k-strip cover of (w_{t-1}, β_{t-1}) for every t. The *(p,k)-Pieri problem* for u asks to enumerate every admissible k-chain from u of length at most p. These chains index the permutations occurring in $e_p(x_1,\dots,x_k)\ \mathfrak{S}_u$, where e_p is the elementary symmetric polynomial.

Method 4.6 (**pieri**(u,p,k))**.** Given $u \in S_n$, $1 \le k < n$, and $0 \le p \le k$, the algorithm returns a list T of pairs (w,t), one for each admissible k-chain from u of length $t \le p$, recording its endpoint w.

(1) Initialize $T \leftarrow [(u,0)]$ and a frontier $F \leftarrow [(u,\infty)]$.
(2) For $t = 1,\dots,p$, form a new frontier $F' \leftarrow [\,]$ and, for each $(w,\beta) \in F$:
   (a) compute $A \leftarrow \{\, a \in \{1,\dots,k\} \mid w(a) < \beta \,\}$;
   (b) for each $b = k+1,\dots,n$ with $w(b) < \beta$, and each $a \in A$ for which t_{ab} is a k-strip cover of (w,β), append $(wt_{ab},\, w(b))$ to F' and $(wt_{ab},\, t)$ to T.
   Then set $F \leftarrow F'$.
(3) Return T.

Lemma 4.7 (Multiplicity-freeness)**.** *For each t, distinct admissible k-chains from u of length t have distinct endpoints; equivalently, the endpoint map is a bijection from admissible k-chains of length t onto the permutations w with $w \geqslant^k u$ of length $\ell(u)+t$ produced by the Pieri rule for $e_t(x_1,\dots,x_k)$.*

Proof. The strictly decreasing bounds $\beta_0 > \beta_1 > \cdots$ force the values moved leftward across position k to be strictly decreasing, which is exactly the condition making the e_t-Pieri expansion multiplicity-free; see [2]. Reconstructing the chain from its endpoint by peeling off these moves in decreasing order of value shows the map is injective, hence bijective onto the Pieri support. □

We now describe the v-side primitive. Its starting point is the structure of a *dominant* double Schubert polynomial. If μ is dominant with $\mathfrak{c}(\mu) = (\lambda_1,\dots,\lambda_m)$, then $\mathfrak{S}_\mu(x;z)$ factors as a product of factorial elementary symmetric polynomials of maximal degree, one per column of the shape,

$$\mathfrak{S}_\mu(x;z) = \prod_{i=1}^{m} E_{\lambda_i,\,\lambda_i}(x; z^{(i)}),$$

where $z^{(i)}$ is the shifted z-alphabet attached to column i (Definition 3.1). For a general v one writes $\mathfrak{S}_v(x;z)$ by applying divided differences in the z-variables to the dominant polynomial $\mathfrak{S}_{\mu(v^{-1})}(x;z)$; because that polynomial factors by column, the divided-difference action decouples and may be carried out one column (one $z^{(i)}$-group) at a time. Applied to a single column this produces a *signed* descending expansion. The following primitive carries it out.

Definition 4.8 (k-strip descent beneath an edge, the down-Pieri problem)**.** Let $w, \mu \in S_n$ and $1 \le k < n$. Say w' lies *beneath the edge* μ if

$$\mathrm{inv}(w'\mu) = \mathrm{inv}(\mu) - \mathrm{inv}(w'),$$

that is, $\ell(w'\mu) = \ell(\mu) - \ell(w')$. A *$k$-strip descent* of w is a Bruhat descent $w \gtrdot wt_{ab}$ realized by a transposition t_{ab} that moves the value in the pivot position k either leftward (from a position $a < k$, with sign -1) or rightward (to a position $b > k$, with sign $+1$), touching only positions still holding their original value. Given a starting label w, the edge μ, a degree p, and the column k, the *down-Pieri problem* asks for the signed list of those length-t descending k-strip chains from w ($t \le p$) whose endpoint lies beneath μ.

Method 4.9 (**kdown**(w, μ, p, k))**.** The algorithm returns a signed list D of triples (w', t, s), one for each length-t descending k-strip chain from w ($t \le p$) whose endpoint w' lies beneath μ, with $s \in \{+1, -1\}$ its sign.

(1) Initialize $D \leftarrow [\,]$; if w is already beneath μ, append $(w, 0, +1)$. Set a frontier $F \leftarrow [(w, +1)]$.
(2) For $t = 1, \dots, p$, form $F' \leftarrow [\,]$ and, for each $(w'', s) \in F$:
   (a) (leftward) for each position $a < k$ with $w''(a) = w(a)$ such that $w'' \gtrdot w''t_{a,k}$ is a Bruhat descent, set $w''' \leftarrow w''t_{a,k}$ and append $(w''', -s)$ to F'; if w''' is beneath μ, append $(w''', t, -s)$ to D;
   (b) (rightward) for each position $b > k$ with $w''(b) = w(b)$ such that $w'' \gtrdot w''t_{k,b}$ is a Bruhat descent, set $w''' \leftarrow w''t_{k,b}$ and append $(w''', +s)$ to F'; if w''' is beneath μ, append $(w''', t, +s)$ to D.

   Then set $F \leftarrow F'$.
(3) Return D.

Applied without the edge μ, the signed descending expansion produces many extraneous terms that ultimately cancel. The edge is exactly the bound that discards them: only endpoints beneath μ are retained in D. In the assembled v-side (Method 4.10 below), only these retained labels seed the next column, so each stage passes forward only labels compatible with the remaining shape.

Method 4.10 (**vpaths**(v), the v-side path dictionaries)**.** Let $\theta := \theta(v^{-1})$ with nonzero parts in columns $1, \dots, L$, and let $\nu := v\,\mu(v^{-1})$. Processing columns from $i = L$ down to $i = 1$, maintain for each column a set of live labels (initialized at column L to the single label ν). At column i, with $k = i$ and edge $\mu_i := \mu$ of the shape truncated below column i, call **kdown**(w, μ_i, θ_i, k) for each live label w; record its signed output triples as the transitions out of w, and let their endpoints be the live labels seeding column $i - 1$. The collected transitions form the path dictionaries used by **schubmult**.

Lemma 4.11 (Correctness of the edge filter)**.** *A permutation w' produced by a descending k-strip chain from w is retained by* **kdown**(w, μ, p, k) *if and only if w' lies beneath μ, i.e.* $\mathrm{inv}(w'\mu) = \mathrm{inv}(\mu) - \mathrm{inv}(w')$*. In the assembly of Method 4.10 the retained signed labels are exactly the terms of $\mathfrak{S}_v(x; z)$ contributed through column i, so no extraneous (cancelling) label propagates to column $i - 1$.*

Proof. The retention test is the identity $\mathrm{inv}(w'\mu) = \mathrm{inv}(\mu) - \mathrm{inv}(w')$ checked in step (2); this is precisely the condition that $w' \le \mu$ with $\ell(w'\mu) = \ell(\mu) - \ell(w')$. The factorization of the dominant polynomial by column, together with the column-wise decoupling of the z-divided differences, identifies the beneath-μ endpoints at column i with the surviving monomials of $\mathfrak{S}_v(x; z)$ after processing that column; the extraneous chains are exactly those leaving the shape, which the edge discards before they seed column $i - 1$. □

Method 4.12 (The algorithm **schubmult**)**.** Fix n and let $u, v \in S_n$. The algorithm **schubmult**(u, v) computes the expansion of $\mathfrak{S}_u(x; z)\ \mathfrak{S}_v(x; z)$ in the Schubert basis as follows. Set $\theta := \theta(v^{-1})$ and let

$$L := \#\{i \mid \theta_i \neq 0\}$$

be the number of nonzero parts. The algorithm proceeds in L layers. It maintains a *front*, a collection of nodes each consisting of an intermediate permutation on the u-side paired with a label on the v-side, initialized to the single node $(u,\ e)$. At layer i it replaces each node by the nodes obtained by combining one u-side cover, generated by **pieri**$(\,\cdot\,, \theta_i, \theta_i)$ (Method 4.6), with one v-side transition, generated by **vpaths**(v) (Method 4.10), accumulating coefficients in the coefficient ring. After all L layers the surviving coefficients at v-label $v\mu(v^{-1})$ give the structure constants.

Quantum version. The same path organization gives the quantum multiplication algorithm. Its layers are indexed by the medium theta $\overline{\theta}(v^{-1})$; equal consecutive parts that are retained in this shape are handled together by the paired quantum Pieri step. On the u-side, classical covers are replaced by the quantum covers of Remark 2.2, including length-decreasing moves, whose contributions carry the corresponding q-monomials. The v-side continues to use the descending path dictionaries of Method 4.10. Thus the two algorithms share the same layer-by-layer structure, while their Pieri transitions reflect the classical or quantum multiplication rule, respectively.

The algorithmic construction is one way to compute products. We now turn to a complementary operator description, which packages the quantum multiplication rules in the Fomin–Gelfand–Postnikov difference operators.

4.1. **Difference operators and multiplication coefficients.**

Definition 4.13 ($\chi_{j;n}$, $\chi^{(n)}$)**.** Recall the commuting difference operators $\chi_{j;n}$, $j \leq n$, of Fomin, Gelfand, and Postnikov [5], acting on polynomials in $x_1, \ldots, x_n$ (with coefficients in any further variables):

$$\chi_{j;n} = x_j - \sum_{1 \leq i < j} q_{ij} \partial^{t_{ij}} + \sum_{j < k \leq n} q_{jk} \partial^{t_{jk}},$$

where for a transposition t we write ∂^t for the divided difference operator ∂_t in the x-variables associated with any reduced word of t, so that $\partial^t \mathfrak{S}_v(x;z) = \mathfrak{S}_{vt}(x;z)$ if $\ell(vt) = \ell(v) - \ell(t)$ and $\partial^t \mathfrak{S}_v(x;z) = 0$ otherwise. We denote the sequence $(\chi_{1;n}, \ldots, \chi_{n;n})$ with n fixed by $\chi^{(n)}$. Fomin, Gelfand and Postnikov proved [5] that if $v \in S_n$ then

$$\widetilde{\mathfrak{S}}_v(\chi^{(n)})(1) = \mathfrak{S}_v(x),$$

and hence, by Lemma 2.1 with $y = 0$ (whose coefficients $\mathfrak{S}_{v_1}(0;z)$ commute with the operators),

$$\widetilde{\mathfrak{S}}_v(\chi^{(n)};z)(1) = \sum_{v_1 \cdot v_2 = v} \mathfrak{S}_{v_1}(0;z)\, \mathfrak{S}_{v_2}(x) = \mathfrak{S}_v(x;z).$$

We specify a more precise formula, as follows.

Theorem 4.14. *Suppose $v \in S_n$. Then*

$$\widetilde{\mathfrak{S}}_v(\chi^{(n)};z) = \sum_{u \in S_n, d} q^d c^{1,d}_{u,v}(x;z) \partial^u$$

Proof. Note that if $\mathfrak{S}_u(x)$ has at most n variables, then

$$\widetilde{\mathfrak{S}}_u(\chi^{(n)};y)\, \widetilde{\mathfrak{S}}_v(\chi^{(n)};z) = \sum_{w,d} q^d c^{w,d}_{u,v}(y;z)\, \widetilde{\mathfrak{S}}_w(\chi^{(n)};y)$$

We thus have that

$$(\widetilde{\mathfrak{S}}_u(\chi^{(n)};y)\, \widetilde{\mathfrak{S}}_v(\chi^{(n)};z))(1) = \sum_{w,d} q^d c^{w,d}_{u,v}(y;z)\, \mathfrak{S}_w(x;y)$$

This is also equal to

$$\widetilde{\mathfrak{S}}_v(\chi^{(n)};z)(\mathfrak{S}_u(x;y))$$

due to the fact that $(\widetilde{\mathfrak{S}}_u(\chi^{(n)};y))(1) = \mathfrak{S}_u(x;y)$. To obtain the coefficient of ∂^u in $\widetilde{\mathfrak{S}}_v(\chi^{(n)};z)$, we set $y = x$ after applying this operator. Since $\mathfrak{S}_w(x;x) = 0$ unless $w = 1$, we have that

$$\widetilde{\mathfrak{S}}_v(\chi^{(n)};z)(\mathfrak{S}_u(x;y))|_{y=x} = \sum_d q^d c^{1,d}_{u,v}(x;z)$$

and the result follows. □

Theorem 4.15. *Let k be an integer and let $v, w \in S_\infty$. Then*

$$\sum_d q^d c^{w,d}_{s_k,v}(y;z) = c^w_{s_k,v}(y;z) + \sum_{i \leq k < j} q_{ij}\, \mathfrak{S}_{vt_{ij}w^{-1}}(y;z),$$

where the sum is over those $i < j$ with $\ell(vt_{ij}) = \ell(v) - \ell(t_{ij})$, and $\mathfrak{S}_{vt_{ij}w^{-1}}(y;z)$ is to be read as 0 unless $\ell(vt_{ij}) = \ell(vt_{ij}w^{-1}) + \ell(w)$.

Proof. We have that the desired value is equal to the coefficient of $\mathfrak{S}_w(x;y)$ in

$$\widetilde{\mathfrak{S}}_{s_k}(\chi^{(n)};y)(\mathfrak{S}_v(x;z))$$

We have that

$$\widetilde{\mathfrak{S}}_{s_k}(\chi^{(n)};y) = \mathfrak{S}_{s_k}(x;y) + \sum_{i \leq k < j \leq n} q_{ij} \partial^{t_{ij}}$$

Applying this operator, we obtain

$$\sum_{i \leq k < j \leq n} q_{ij}\, \mathfrak{S}_{vt_{ij}}(x;z) + \sum_w c^w_{s_k,v}(y;z)\, \mathfrak{S}_w(x;y)$$

Applying the Cauchy formula (Lemma 2.1 at $q = 0$), this is equal to

$$\sum_{i \le k < j} \sum_{u} q_{ij}\, \mathfrak{S}_{vt_{ij}u^{-1}}(y;z)\, \mathfrak{S}_u(x;y) + \sum_{w} c^{w}_{s_k,v}(y;z)\, \mathfrak{S}_w(x;y),$$

the inner sum being over the reduced factorizations $vt_{ij} = (vt_{ij}u^{-1}) \cdot u$. Setting $u = w$ and allowing n to be sufficiently large, we obtain the result. □

5. Quantum to classical reduction

In this section we extend the quantum-to-classical reduction of Huang and Li [7] from the equivariant quantum structure constants $c^{w,d}_{u,v}(y;y)$ to the triple constants. We write $\mathrm{sgn}_i(w) = 1$ if $w(i) > w(i+1)$ and $\mathrm{sgn}_i(w) = 0$ otherwise, $\delta_i \in Q^\vee$ for the weak composition with a single 1 in position i, and $|d| = \sum_i d_i$; the bracket $\langle d \rangle_i = 2d_i - d_{i-1} - d_{i+1}$ was defined in Section 2. We say that $u \in S_\infty$ is *k-Grassmannian* if k is its only descent.

Theorem 5.1 ([7, Theorem 2.7])**.** *We have the following.*

(1) $c^{w,d}_{u,v}(y;y) = 0$ *unless* $\mathrm{sgn}_i(w) + \langle d \rangle_i \le \mathrm{sgn}_i(u) + \mathrm{sgn}_i(v)$ *for all* i.
(2) *If* $\mathrm{sgn}_k(w) + \langle d \rangle_k = \mathrm{sgn}_k(u) + \mathrm{sgn}_k(v) = 2$ *for some* k, *then*

$$c^{w,d}_{u,v}(y;y) = c^{ws_k,d-\delta_k}_{us_k,vs_k}(y;y) = \begin{cases} c^{ws_k,d-\delta_k}_{us_k,v}(y;y) & \text{if } \mathrm{sgn}_k(w) = 0 \\ c^{ws_k,d}_{us_k,v}(y;y) & \text{if } \mathrm{sgn}_k(w) = 1 \end{cases}$$

This is symmetric in u and v.

Corollary 5.2 ([7, Corollary 2.8])**.** *Let $u, v, w \in S_\infty$ and $d \in Q^\vee$. Fix an index i.*

(1) *If* $\langle d \rangle_i = \mathrm{sgn}_i(u) = 0$ *and* $\mathrm{sgn}_i(w) = \mathrm{sgn}_i(v) = 1$, *then*

$$c^{w,d}_{u,v}(y;y) = c^{ws_i,d}_{u,vs_i}(y;y).$$

(2) *If* $\langle d \rangle_i = \mathrm{sgn}_i(u) = 1$ *and* $\mathrm{sgn}_i(w) = \mathrm{sgn}_i(v) = 0$, *then*

$$c^{w,d}_{u,v}(y;y) = c^{ws_i,d-\delta_i}_{us_i,v}(y;y).$$

This is symmetric in u and v.

Theorem 5.3 (Quantum reduction holds for triple quantum Schubert coefficients)**.** *We have the following.*

(1) $c^{w,d}_{u,v}(y;z) = 0$ *unless* $\mathrm{sgn}_i(w) + \langle d \rangle_i \le \mathrm{sgn}_i(u) + \mathrm{sgn}_i(v)$ *for all* i.
(2) *If* $\mathrm{sgn}_k(w) + \langle d \rangle_k = \mathrm{sgn}_k(u) + \mathrm{sgn}_k(v) = 2$ *for some* k, *then*

$$c^{w,d}_{u,v}(y;z) = c^{ws_k,d-\delta_k}_{us_k,vs_k}(y;z) = \begin{cases} c^{ws_k,d-\delta_k}_{us_k,v}(y;z) & \text{if } \mathrm{sgn}_k(w) = 0 \\ c^{ws_k,d}_{us_k,v}(y;z) & \text{if } \mathrm{sgn}_k(w) = 1 \end{cases}$$

This is symmetric in u and v.

Proof. For (1), we note that since $\mathrm{sgn}_i(v_2) \le \mathrm{sgn}_i(v)$ for any factorization $v_1 \cdot v_2 = v$, we have that $\mathrm{sgn}_i(w) + \langle d \rangle_i \le \mathrm{sgn}_i(u) + \mathrm{sgn}_i(v_2)$ is a necessary condition for $c^{w,d}_{u,v_2}(y;y) \neq 0$. The result then follows from Lemma 2.1.

For (2), we note that if $\mathrm{sgn}_k(w) + \langle d \rangle_k = \mathrm{sgn}_k(u) + \mathrm{sgn}_k(v) = 2$, then since $\mathrm{sgn}_k(v_2) = \mathrm{sgn}_k(v)$ for any factorization $v_1 \cdot v_2 = v$ such that $c^{w,d}_{u,v_2}(y;y) \neq 0$, Theorem 5.1(2) holds for each term in the sum in Lemma 2.1, and the result follows. □

Corollary 5.4 (z-version of Corollary 5.2(2))**.** *Let $u, v, w \in S_\infty$ and $d \in Q^\vee$, and fix an index i with* $\mathrm{sgn}_i(v) = 0$. *If* $\langle d \rangle_i = \mathrm{sgn}_i(u) = 1$ *and* $\mathrm{sgn}_i(w) = 0$, *then*

$$c^{w,d}_{u,v}(y;z) = c^{ws_i,d-\delta_i}_{us_i,v}(y;z).$$

Proof. Since $\mathrm{sgn}_i(v_2) \le \mathrm{sgn}_i(v) = 0$ for every factorization $v_1 \cdot v_2 = v$, the hypotheses of Corollary 5.2(2) hold for each term $c^{w,d}_{u,v_2}(y;y)$ appearing in Lemma 2.1, so $c^{w,d}_{u,v_2}(y;y) = c^{ws_i,d-\delta_i}_{us_i,v_2}(y;y)$. Resumming via Lemma 2.1 gives the claim. □

Lemma 5.5 (Reduction to a coroot supported at k)**.** *Let $u, v, w \in S_\infty$ and let $d \in Q^\vee$. Suppose v has precisely one descent, which is k. If $c_{u,v}^{w,d}(y;z) \neq 0$, then there exist $u', w' \in S_\infty$ and $d' \in Q^\vee$ with $0 \leq d' \leq d$ such that*

$$c_{u,v}^{w,d}(y;z) = c_{u',v}^{w',d'}(y;z)$$

and $\langle d' \rangle_i \leq 0$ for all $i \neq k$. Moreover, if $d' \neq 0$ then $\langle d' \rangle_k > 0$.

Proof. Suppose $\langle d \rangle_i > 0$ for some $i \neq k$, and choose such an i to be maximal. Because v is k-Grassmannian, $\mathrm{sgn}_i(v) = 0$, so Theorem 5.3(1) forces

$$\mathrm{sgn}_i(w) + \langle d \rangle_i \leq \mathrm{sgn}_i(u) + \mathrm{sgn}_i(v) = \mathrm{sgn}_i(u) \leq 1.$$

Since $\langle d \rangle_i \geq 1$, we conclude $\langle d \rangle_i = 1$, $\mathrm{sgn}_i(u) = 1$, and $\mathrm{sgn}_i(w) = 0$. These are exactly the hypotheses of Corollary 5.4 (note $\mathrm{sgn}_i(v) = 0$), so

$$c_{u,v}^{w,d}(y;z) = c_{us_i,v}^{ws_i,d-\delta_i}(y;z).$$

Moreover $\langle d \rangle_i = 2d_i - d_{i-1} - d_{i+1} = 1$ with $d_{i\pm1} \geq 0$ forces $d_i \geq 1$, so $d - \delta_i \in Q^\vee$ with $0 \leq d - \delta_i \leq d$ and $|d - \delta_i| < |d|$. Replacing (u, w, d) by $(us_i, v \text{ fixed}, ws_i, d - \delta_i)$ and iterating, the quantity $|d|$ strictly decreases at each step, so the process terminates in a triple (u', w', d') with $\langle d' \rangle_i \leq 0$ for all $i \neq k$.

Finally, for any finitely supported $d' \in Q^\vee$ the values $\langle d' \rangle_i = 2d'_i - d'_{i-1} - d'_{i+1}$ telescope to $\sum_i \langle d' \rangle_i = 0$. Hence $\langle d' \rangle_k = -\sum_{i \neq k} \langle d' \rangle_i \geq 0$, with equality if and only if $\langle d' \rangle_i = 0$ for all i, i.e. $d' = 0$. Thus $d' \neq 0$ implies $\langle d' \rangle_k > 0$. □

6. Proof of the Pieri formula

Throughout this section permutations act on positions: $(vt_{ab})(a) = v(b)$, $(vt_{ab})(b) = v(a)$, and $(vt_{ab})(r) = v(r)$ otherwise. A cycle is written $(a_p, \ldots, a_1, b)$ with the meaning $a_p \mapsto a_{p-1} \mapsto \cdots \mapsto a_1 \mapsto b \mapsto a_p$; when $b = \max\{a_1, \ldots, a_p, b\}$ we call b the *top* of the cycle and refer to

$$(a_p, \ldots, a_1, b) = t_{a_1 b}\, t_{a_2 b} \cdots t_{a_p b} \tag{6.1}$$

as its *pivot factorization*. (Indeed, the right-hand side sends $a_p \mapsto b \mapsto a_{p-1}$, $b \mapsto a_p$ and $a_1 \mapsto b$.)

6.1. The single-variable Monk formula.

Lemma 6.1. *Let $u \in S_\infty$ and let $i, j \geq 1$. Then*

$$(x_i - z_j)\,\widetilde{\mathfrak{S}}_u(x;y) = (y_{u(i)} - z_j)\,\widetilde{\mathfrak{S}}_u(x;y) + \sum_{\substack{p \neq i \\ \ell(ut_{ip}) = \ell(u)+1}} \mathrm{sign}(p-i)\,\widetilde{\mathfrak{S}}_{ut_{ip}}(x;y)$$

$$+ \sum_{\substack{p \neq i \\ \ell(ut_{ip}) = \ell(u) - 2|p-i|+1}} \mathrm{sign}(p-i) q_{ip}\,\widetilde{\mathfrak{S}}_{ut_{ip}}(x;y)$$

where q_{ip} stands for $q_{\min(i,p)\max(i,p)}$.

Proof. The quantum double Schubert polynomials represent the Schubert classes in the torus-equivariant quantum cohomology of the flag manifold [10], so they satisfy the equivariant quantum Chevalley formula of Mihalcea [13, Theorem 1]. In type A, with $\widetilde{\mathfrak{S}}_{s_m}(x;y) = \sum_{a \leq m}(x_a - y_a)$, that formula reads

$$\widetilde{\mathfrak{S}}_{s_m}(x;y)\,\widetilde{\mathfrak{S}}_u(x;y) = \sum_{a \leq m} (y_{u(a)} - y_a)\,\widetilde{\mathfrak{S}}_u(x;y) + \sum_{\substack{a \leq m < b \\ \ell(ut_{ab}) = \ell(u)+1}} \widetilde{\mathfrak{S}}_{ut_{ab}}(x;y) + \sum_{\substack{a \leq m < b \\ \ell(ut_{ab}) = \ell(u) - 2(b-a)+1}} q_{ab}\,\widetilde{\mathfrak{S}}_{ut_{ab}}(x;y),$$

the diagonal coefficient being the localization of $\widetilde{\mathfrak{S}}_{s_m}(x;y)$ at u, and the two sums ranging over the classical and quantum covers of u in the quantum Bruhat graph (for $\alpha = e_a - e_b$ one has $\langle \omega_m, \alpha^\vee \rangle = 1$ exactly when $a \leq m < b$, and $\langle 2\rho, \alpha^\vee \rangle = 2(b-a)$). By stability we may work in S_n for n large. Now

$$x_i - z_j = \widetilde{\mathfrak{S}}_{s_i}(x;y) - \widetilde{\mathfrak{S}}_{s_{i-1}}(x;y) + y_i - z_j,$$

with the term $\widetilde{\mathfrak{S}}_{s_{i-1}}$ omitted when $i = 1$. Subtracting the two Chevalley formulas, the diagonal coefficients combine to $(y_{u(i)} - y_i) + (y_i - z_j) = y_{u(i)} - z_j$, and the reflections t_{ab} with $a < i < b$ occur in both sums with the same coefficient and cancel. What remains are the reflections t_{ib} with $b > i$ (from $\widetilde{\mathfrak{S}}_{s_i}$, with sign $+1$) and t_{ai} with $a < i$ (from $\widetilde{\mathfrak{S}}_{s_{i-1}}$, with sign -1), each carrying the factor 1 or q_{ab} according to whether the cover is classical or quantum. This is the asserted formula. □

6.2. **Covers and tight cycles.** The following classical count is the only fact about lengths that we shall use.

Lemma 6.2. *Let $v \in S_\infty$ and $a < b$. If $v(a) < v(b)$ then*

$$\ell(vt_{ab}) = \ell(v) + 1 + 2\,\#\{r \mid a < r < b,\ v(a) < v(r) < v(b)\},$$

and if $v(a) > v(b)$ then

$$\ell(vt_{ab}) = \ell(v) - 1 - 2\,\#\{r \mid a < r < b,\ v(b) < v(r) < v(a)\}.$$

In particular $\ell(vt_{ab}) \geq \ell(v) + 1$ if $v(a) < v(b)$, and $\ell(vt_{ab}) \geq \ell(v) - (2(b-a) - 1)$ always.

Proof. Passing from v to vt_{ab} changes the inversion status of the pair of positions (a, b), and of the pairs (a, r), (r, b) with $a < r < b$ exactly when $v(r)$ lies strictly between $v(a)$ and $v(b)$ (in which case both pairs change); all other pairs are unaffected. □

Definition 6.3 (Covers, $\lhd$, κ)**.** For $v \in S_\infty$ and $a < b$ we write $v \lhd vt_{ab}$, and say that t_{ab} is a *cover* of v, if either $\ell(vt_{ab}) = \ell(v) + 1$ (a *classical* cover) or $\ell(vt_{ab}) = \ell(v) - 2(b-a) + 1$ (a *quantum* cover). We set $\kappa(v, t_{ab}) = 1$ in the first case and $\kappa(v, t_{ab}) = q_{ab}$ in the second. By Lemma 6.2, t_{ab} is a classical cover of v if and only if $v(a) < v(b)$ and no r with $a < r < b$ satisfies $v(a) < v(r) < v(b)$, and a quantum cover if and only if $v(a) > v(b)$ and every r with $a < r < b$ satisfies $v(b) < v(r) < v(a)$. These are the covers of the quantum Bruhat graph, and condition (4) of Definition 3.1 says precisely that each step of a witnessing sequence is a cover (Remark 2.2).

It is convenient to separate the two values being exchanged from the values lying between them. For distinct numbers $\alpha \neq \beta$ and a finite set J of numbers not containing α, β, say that the pair (α, β) is *J-admissible* if either $\alpha < \beta$ and $J \cap (\alpha, \beta) = \emptyset$, or $\alpha > \beta$ and $J \subseteq (\beta, \alpha)$. Thus t_{ab} is a cover of v if and only if $(v(a), v(b))$ is J-admissible for $J = \{v(r) \mid a < r < b\}$; the cover is classical when $\alpha < \beta$ and quantum when $\alpha > \beta$.

Lemma 6.4. *Let α, β, γ be distinct numbers and I, J, J' finite sets of numbers avoiding them.*

(i) *(α, β) is $(J \cup J')$-admissible if and only if it is J-admissible and J'-admissible. In particular $J' \subseteq J$ and J-admissibility imply J'-admissibility.*
(ii) *(Composition.) If $(\alpha_1, \alpha_0), (\alpha_2, \alpha_1), \ldots, (\alpha_n, \alpha_{n-1})$ are I-admissible then so is (α_n, α_0).*
(iii) *(Triangle.) If (γ, β) is I-admissible and (α, γ) is $(J \cup \{\beta\} \cup I)$-admissible, then (α, β) is I-admissible and (α, γ) is J-admissible.*

Proof. (i) is immediate from the definition. For (ii), if $I = \emptyset$ there is nothing to prove. Otherwise I cuts the complement of I into intervals ("blocks"), among them $V_- = (-\infty, \min I)$ and $V_+ = (\max I, \infty)$. An I-admissible pair (α', α) with $\alpha' < \alpha$ has α, α' in the same block, and one with $\alpha' > \alpha$ has $\alpha \in V_-$ and $\alpha' \in V_+$. Hence along the chain the values descend within a block, except for at most one jump from V_- to V_+, after which they remain in V_+. Either $\alpha_n < \alpha_0$ lie in a common block, or $\alpha_0 \in V_-$ and $\alpha_n \in V_+$; in both cases (α_n, α_0) is I-admissible. For (iii), the second assertion is (i). For the first there are four cases. If $\gamma < \beta$ and $\alpha < \gamma$: then $\alpha < \beta$ and $I \cap (\alpha, \beta) \subseteq (I \cap (\alpha, \gamma)) \cup (I \cap [\gamma, \beta)) = \emptyset$. If $\gamma < \beta$ and $\alpha > \gamma$: then $\beta \in (\gamma, \alpha)$ gives $\beta < \alpha$, and $I \subseteq (\gamma, \alpha)$ together with $I \cap (\gamma, \beta) = \emptyset$ gives $I \subseteq (\beta, \alpha)$. If $\gamma > \beta$ and $\alpha < \gamma$: then $\beta \notin (\alpha, \gamma)$ forces $\beta < \alpha$, and $I \subseteq (\beta, \gamma)$ with $I \cap (\alpha, \gamma) = \emptyset$ gives $I \subseteq (\beta, \alpha)$. The case $\gamma > \beta$, $\alpha > \gamma$ cannot occur, since it would require $\beta \in (\gamma, \alpha)$. □

Definition 6.5 (Descents of a cycle, $\mathrm{desc}_u(c)$)**.** Let c be a cycle and suppose $u \in S_\infty$. Write c as

$$(a_1, \ldots, a_n)$$

where $a_n > a_i$ for all $i < n$ (this is a unique way to express the cycle since the a_j are distinct). Then for any i with $1 \leq i < n$ we say that a_i is a *descent* of c with respect to u if $u(a_i) > u(a_{i+1})$. We denote the set of all descents of c with respect to u by $\mathrm{desc}_u(c)$.

In the notation $c = (a_p, \ldots, a_1, b)$ of (6.1), with $a_0 := b$, we have $a_j \in \mathrm{desc}_u(c)$ if and only if $u(a_j) > u(a_{j-1})$; the top b is never a descent.

Definition 6.6 ($L_u(\sigma)$, $\mathbf{q}_u(\sigma)$, m-shaped, u-tight)**.** Let $u, \sigma \in S_\infty$ and let $c_1, \dots, c_n$ be the nontrivial cycles of σ, with $c_i = (a_{i,p_i}, \dots, a_{i,1}, b_i)$ and $b_i = \max c_i$. Put

$$L_u(\sigma) = \sum_{i=1}^{n} \Big(p_i - 2 \sum_{a \in \mathrm{desc}_u(c_i)} (b_i - a) \Big), \qquad \mathbf{q}_u(\sigma) = \prod_{i=1}^{n} \prod_{a \in \mathrm{desc}_u(c_i)} q_{ab_i}.$$

Both depend only on the restriction of u to $\mathrm{supp}(\sigma)$, and $\deg \mathbf{q}_u(\sigma) = \sum_i p_i - L_u(\sigma)$. We say that σ is *m-shaped* if every nontrivial cycle of σ contains exactly one element greater than m, and that σ is *u-tight* if

$$\ell(u\sigma) - \ell(u) = L_u(\sigma).$$

A *pivot chain* of σ is a sequence $t_1, \dots, t_N$ ($N = \sum_i p_i$) of transpositions obtained by interleaving the pivot factorizations (6.1) of $c_1, \dots, c_n$ in any manner that preserves the order of the factors within each cycle; since the cycles are disjoint, $t_1 \cdots t_N = \sigma$.

Lemma 6.7 (Tightness)**.** *Let $u, \sigma \in S_\infty$, let $t_1, \dots, t_N$ be a pivot chain of σ, and put $u_0 = u$, $u_j = u_{j-1} t_j$.*

(i) *If $t_j = t_{a_{i,l} b_i}$ then $u_{j-1}(a_{i,l}) = u(a_{i,l})$ and $u_{j-1}(b_i) = u(a_{i,l-1})$ (with $a_{i,0} = b_i$). Thus the j-th step exchanges the values $u(a_{i,l})$ and $u(a_{i,l-1})$, and $a_{i,l} \in \mathrm{desc}_u(c_i)$ exactly when the larger of the two sits at the larger position b_i before the step.*

(ii) *$\ell(u\sigma) - \ell(u) \geq L_u(\sigma)$.*

(iii) *σ is u-tight if and only if $u_{j-1} \lhd u_j$ for every j. In that case the j-th step is a classical cover if $a_{i,l} \notin \mathrm{desc}_u(c_i)$ and a quantum cover if $a_{i,l} \in \mathrm{desc}_u(c_i)$, and $\prod_j \kappa(u_{j-1}, t_j) = \mathbf{q}_u(\sigma)$.*

(iv) *If $\sigma = \rho\tau$ where ρ and τ are the products of two complementary sets of cycles of σ, then σ is u-tight if and only if ρ is u-tight and τ is $u\rho$-tight. Moreover $L_u(\tau) = L_{u\rho}(\tau)$ and $\mathbf{q}_u(\tau) = \mathbf{q}_{u\rho}(\tau)$.*

(v) *If σ is u-tight then so is $t_1 \cdots t_j$ for every j, and u_j is obtained from u by a sequence of covers.*

Proof. (i) The position $a_{i,l}$ is moved only by the factor $t_{a_{i,l} b_i}$, so before that factor is applied it carries $u(a_{i,l})$; the position b_i is moved only by the factors of c_i, and after $t_{a_{i,l-1} b_i}$ it carries $u(a_{i,l-1})$ (and $u(b_i)$ before any factor of c_i). For (ii) and (iii), by (i) and Lemma 6.2 the j-th step increases the length by at least 1 if $a_{i,l} \notin \mathrm{desc}_u(c_i)$ and by at least $-(2(b_i - a_{i,l}) - 1)$ if $a_{i,l} \in \mathrm{desc}_u(c_i)$; in both cases equality holds exactly when t_j is a cover of u_{j-1}, classical in the first case and quantum in the second (a step with $u_{j-1}(a_{i,l}) < u_{j-1}(b_i)$ increases the length and so cannot be a quantum cover, and symmetrically). Summing over j gives (ii), and shows that equality in (ii) holds exactly when every step is a cover; the product of the κ's is then the product of $q_{a_{i,l} b_i}$ over the descents, which is $\mathbf{q}_u(\sigma)$. For (iv), $\ell(u\rho\tau) - \ell(u) = (\ell(u\rho) - \ell(u)) + (\ell(u\rho\tau) - \ell(u\rho)) \geq L_u(\rho) + L_{u\rho}(\tau) = L_u(\rho) + L_u(\tau) = L_u(\sigma)$, where $L_{u\rho}(\tau) = L_u(\tau)$ because $u\rho$ and u agree on $\mathrm{supp}(\tau)$; equality holds if and only if both inequalities are equalities. (v) The partial product $t_1 \cdots t_j$ is again a product of disjoint cycles of the form $(a_{i,l}, \dots, a_{i,1}, b_i)$ (a cycle of σ truncated after its first l factors), and $t_1, \dots, t_j$ is a pivot chain of it, all of whose steps are covers by (iii). □

With this language the cycle description of $\overset{k}{\rightsquigarrow}$ becomes a statement in which the chain disappears entirely.

Lemma 6.8. *Suppose $u, w \in S_\infty$ and let $k \geq 0$ be an integer, and put $\pi = u^{-1} w$. Then $u \overset{k}{\rightsquigarrow} w$ if and only if π is k-shaped and u-tight. In that case the pivot chain of π in which the cycles are taken in order of decreasing tops is a sequence $t_{a_1 b_1}, \dots, t_{a_p b_p}$ as in Definition 3.1, every pivot chain of π is a chain of covers, and*

$$q_k(u, w) = \mathbf{q}_u(\pi).$$

In particular $q_k(u, w)$ does not depend on the witnessing sequence, and $\ell(u, w) + \deg q_k(u, w)$ equals the sum of the lengths of the cycles of π, each diminished by one.

Proof. Suppose first that $t_{a_1 b_1}, \dots, t_{a_p b_p}$ satisfy conditions (1)–(5) of Definition 3.1. The b_i are weakly decreasing, so the factors sharing a common value b of b_i are consecutive, and by (6.1) their product $t_{a_i b} \cdots t_{a_{i+r} b}$ is the cycle $(a_{i+r}, \dots, a_i, b)$, whose top is b because $a_l \leq k < b$. Cycles belonging to distinct values of b are disjoint, since the a_i are pairwise distinct and no a_i equals any b_j. Hence π is the product of these cycles, each of which contains exactly one element greater than k, namely its top: π is k-shaped. The given sequence is a pivot chain of π, and condition (4) says that each of its steps is a cover; so π is u-tight by Lemma 6.7(iii), and the recursive definition of $q_k(u, w)$ assigns to each step exactly the factor κ, whence $q_k(u, w) = \mathbf{q}_u(\pi)$.

Conversely, suppose π is k-shaped and u-tight. Order the nontrivial cycles of π by decreasing tops and take the corresponding pivot chain. The top of each cycle is its unique element exceeding k, so condition (1) holds;

(2) holds because the cycles are disjoint and the a's within a cycle are distinct; (3) holds by the chosen order; (4) holds by Lemma 6.7(iii); and (5) is $t_1 \cdots t_N = \pi$. The final assertion is $\deg \mathbf{q}_u(\pi) = \sum_i p_i - L_u(\pi)$. □

Corollary 6.9 (Cycle formula for the quantum weight)**.** *Suppose $u \overset{k}{\rightsquigarrow} w$, and write the nontrivial cycles in the decomposition of $u^{-1}w$ as $c_i = (a_{i,p_i}, \dots, a_{i,1}, b_i)$, as in Lemma 6.8. Then*

$$q_k(u,w) = \prod_i \prod_{a \in \mathrm{desc}_u(c_i)} q_{ab_i}.$$

Equivalently, the exponent of q_r in $q_k(u,w)$ is

$$\#\{(i,a) : a \in \mathrm{desc}_u(c_i),\ a \le r < b_i\}.$$

Moreover, $q_k(u,w)$ is multiplicative over the cycles of $u^{-1}w$, and the factor attached to a cycle depends only on the values of u on that cycle.

Proof. This is $q_k(u,w) = \mathbf{q}_u(\pi)$ together with $q_{ab} = q_a \cdots q_{b-1}$. □

We next show that a tight cycle is very constrained: the values of u along it, read cyclically, decrease except at exactly one place. For a cycle $c = (a_p, \dots, a_1, b)$ with top b and a permutation u we write

$$\beta_0 = u(b), \qquad \beta_j = u(a_j) \quad (1 \le j \le p),$$

so that $a_j \in \mathrm{desc}_u(c)$ if and only if $\beta_j > \beta_{j-1}$, and we put $u_j = u\, t_{a_1 b} \cdots t_{a_j b}$. By Lemma 6.7(i), the j-th step of the pivot chain exchanges β_j (at a_j) with β_{j-1} (at b); moreover $u_m(a_l) = \beta_{l-1}$ for $l \le m$ and $u_m(a_l) = \beta_l$ for $l > m$. If c is u-tight, Definition 6.3 gives, for every j and every position r with $a_j < r < b$:

$$\text{if } \beta_j < \beta_{j-1} \text{ (classical step): } u_{j-1}(r) \notin (\beta_j, \beta_{j-1}); \tag{6.2}$$

$$\text{if } \beta_j > \beta_{j-1} \text{ (quantum step): } u_{j-1}(r) \in (\beta_{j-1}, \beta_j). \tag{6.3}$$

Lemma 6.10. *Let $c = (a_p, \dots, a_1, b)$ be a cycle with top b which is u-tight. Then $\mathrm{desc}_u(c)$ has at most one element. In particular this applies to every cycle of $u^{-1}w$ whenever $u \overset{k}{\rightsquigarrow} w$.*

Proof. Suppose $i < j$ are two descents (quantum steps) with no descent strictly between them, so that $\beta_i > \beta_{i-1}$, $\beta_j > \beta_{j-1}$ and $\beta_i > \beta_{i+1} > \cdots > \beta_{j-1}$.

If $a_j > a_i$, then (6.3) for the step i and $r = a_j$ gives $\beta_j = u_{i-1}(a_j) \in (\beta_{i-1}, \beta_i)$, so $\beta_{j-1} < \beta_j < \beta_i$. Hence $j \ge i+2$, and there is m with $i < m < j$ and $\beta_m < \beta_j < \beta_{m-1}$. The step m is classical. If $a_j > a_m$, then (6.2) for the step m and $r = a_j$ is violated by $u_{m-1}(a_j) = \beta_j$. So $a_m > a_j$, and then (6.3) for the step j and $r = a_m$ requires $u_{j-1}(a_m) = \beta_{m-1} \in (\beta_{j-1}, \beta_j)$, contradicting $\beta_{m-1} > \beta_j$.

If $a_j < a_i$, then (6.3) for the step j and $r = a_i$ gives $u_{j-1}(a_i) = \beta_{i-1} \in (\beta_{j-1}, \beta_j)$, so $\beta_{j-1} < \beta_{i-1} < \beta_i$. Hence again $j \ge i+2$ and there is m with $i < m < j$ and $\beta_m < \beta_{i-1} < \beta_{m-1}$. If $a_i > a_m$, (6.2) for the step m and $r = a_i$ is violated by $u_{m-1}(a_i) = \beta_{i-1}$. So $a_m > a_i$, and (6.3) for the step i and $r = a_m$ requires $u_{i-1}(a_m) = \beta_m \in (\beta_{i-1}, \beta_i)$, contradicting $\beta_m < \beta_{i-1}$. □

Definition 6.11 (Redefinition of $\mathrm{desc}_u(c)$)**.** If c is a tight cycle, we redefine $\mathrm{desc}_u(c)$ to be the unique element of the previously defined set $\mathrm{desc}_u(c)$ if it exists, or otherwise 0.

Lemma 6.12. *Let $c = (a_p, \dots, a_1, b)$ be a cycle with top b which is u-tight, and suppose $\mathrm{desc}_u(c) = a_q$ with $q \ge 1$. Then*

$$\beta_q > \beta_{q+1} > \cdots > \beta_p > \beta_0 > \beta_1 > \cdots > \beta_{q-1}.$$

In particular $u(a_j) > u(b)$ for all $j \ge q$.

Proof. All steps other than the q-th are classical, which gives the two decreasing runs; it remains to prove $\beta_l > \beta_0$ for $l \ge q$.

First let $l = q$. For $l' < q$ with $a_{l'} < a_q$, (6.2) for the step l' and $r = a_q$ gives $\beta_q = u_{l'-1}(a_q) \notin (\beta_{l'}, \beta_{l'-1})$. If some $l' < q$ has $a_{l'} > a_q$, let l^* be the least such; then (6.3) for the step q and $r = a_{l^*}$ gives $u_{q-1}(a_{l^*}) = \beta_{l^*-1} \in (\beta_{q-1}, \beta_q)$, so $\beta_q > \beta_{l^*-1}$, and descending induction on $l' < l^*$ using $\beta_q \notin (\beta_{l'}, \beta_{l'-1})$ yields $\beta_q > \beta_{l'-1}$ for all $l' \le l^*$, in particular $\beta_q > \beta_0$. If no such l' exists, the same induction starting from $\beta_q > \beta_{q-1}$ gives $\beta_q > \beta_0$.

Now let $l > q$ and assume $\beta_{l-1} > \beta_0$ (and $\beta_l < \beta_{l-1}$). Suppose, for a contradiction, that $\beta_l < \beta_0$. We claim that $a_{l'} < a_l$ and $\beta_l < \beta_{l'}$ for $1 \le l' \le q-1$. Indeed, assume $\beta_l < \beta_{l'-1}$ (true for $l' = 1$). If $a_{l'} > a_l$, then (6.2)

for the step l and $r = a_{l'}$ requires $u_{l-1}(a_{l'}) = \beta_{l'-1} \notin (\beta_l, \beta_{l-1})$, which is false as $\beta_l < \beta_{l'-1} \le \beta_0 < \beta_{l-1}$. So $a_{l'} < a_l$, and (6.2) for the step l' and $r = a_l$ gives $\beta_l \notin (\beta_{l'}, \beta_{l'-1})$, whence $\beta_l < \beta_{l'}$. The claim yields $\beta_l < \beta_{q-1}$. Finally consider the quantum step q: if $a_q < a_l$, (6.3) with $r = a_l$ requires $\beta_l \in (\beta_{q-1}, \beta_q)$, contradicting $\beta_l < \beta_{q-1}$; if $a_q > a_l$, (6.2) for the step l with $r = a_q$ requires $u_{l-1}(a_q) = \beta_{q-1} \notin (\beta_l, \beta_{l-1})$, contradicting $\beta_l < \beta_{q-1} < \beta_0 < \beta_{l-1}$. Hence $\beta_l > \beta_0$, and the induction is complete. □

Corollary 6.13 (Cyclic monotonicity)**.** *Let $c = (a_p, \dots, a_1, b)$ be a u-tight cycle with top b. Then the cyclic sequence $(\beta_0, \beta_1, \dots, \beta_p)$ has exactly one cyclic ascent: either $\beta_0 > \beta_1 > \cdots > \beta_p$ (and $\mathrm{desc}_u(c) = 0$), or $\mathrm{desc}_u(c) = a_q$ with $q \ge 1$ and $\beta_q > \cdots > \beta_p > \beta_0 > \cdots > \beta_{q-1}$. Consequently*

$$L_u(c) = \begin{cases} p & \text{if the ascent is at the wrap } (\beta_p, \beta_0), \\ p - 2(b - a_q) & \text{if the ascent is } (\beta_{q-1}, \beta_q), \end{cases} \qquad \mathbf{q}_u(c) = \begin{cases} 1, \\ q_{a_q b}. \end{cases}$$

We shall repeatedly use the following elementary observation about such sequences. Call a cyclic sequence of distinct numbers *unimodal* if it has exactly one cyclic ascent, i.e. if it is a cyclic rotation of a decreasing sequence. Every cyclically consecutive block of a unimodal sequence is unimodal, and so is any subsequence. If Γ is unimodal and $\eta \notin \Gamma$, there is exactly one *gap* of Γ (a pair of cyclically consecutive entries) into which η can be inserted so that the result is again unimodal, namely: the descent gap (γ, γ') with $\gamma > \eta > \gamma'$ if $\min\Gamma < \eta < \max\Gamma$, and the ascent gap $(\min\Gamma, \max\Gamma)$ if $\eta < \min\Gamma$ or $\eta > \max\Gamma$. We call it the *insertion gap* of η in Γ.

6.3. **The position k.** We organize the inductive step of Theorem 3.4 around the position k. The following lemma records two facts about the interaction of $\overset{k}{\rightsquigarrow}$ and $\overset{k-1}{\rightsquigarrow}$ at position k.

Lemma 6.14. *Suppose $u, w \in S_\infty$ and let $k > 1$ be an integer.*

(i) *If $u(k) = w(k)$, then $u \overset{k}{\rightsquigarrow} w$ if and only if $u \overset{k-1}{\rightsquigarrow} w$, and in that case $q_k(u, w) = q_{k-1}(u, w)$ and $\mathrm{fix}_k(u, w) = \mathrm{fix}_{k-1}(u, w) \cup \{u(k)\}$.*

(ii) *If $u(k) \ne w(k)$ and $u \overset{k}{\rightsquigarrow} w$, then $u \overset{k-1}{\not\rightsquigarrow} w$.*

Proof. Let $\pi = u^{-1}w$. If $u(k) = w(k)$ then k is a fixed point of π, so no nontrivial cycle of π contains k, and such a cycle has exactly one element greater than k if and only if it has exactly one element greater than $k-1$. Thus π is k-shaped if and only if it is $(k-1)$-shaped, while tightness and $\mathbf{q}_u(\pi)$ do not involve k at all; (i) follows from Lemma 6.8. For (ii), the cycle of π containing k is nontrivial and, π being k-shaped, contains an element greater than k; so it contains two elements greater than $k-1$ and π is not $(k-1)$-shaped. □

For the rest of this section we fix $u, w \in S_\infty$ and $k > 1$, and write

$$\pi = u^{-1}w, \qquad C = \text{the cycle of } \pi \text{ containing } k \ (C = \{k\} \text{ if } \pi(k) = k), \qquad B = \max C.$$

For $m \ge 0$ we set $c_m(u, w) = q_m(u, w)\mathbf{E}(y_{\mathrm{fix}_m(u,w)}; z)$ if $u \overset{m}{\rightsquigarrow} w$ and $c_m(u, w) = 0$ otherwise. The *Monk summands* at w are indexed by the reflections $t \in \{t_{ak} \mid a < k\} \cup \{t_{kb} \mid b > k\}$: writing $v = wt$ and $\sigma = u^{-1}v = \pi t$, the summand of t is

$$\mathrm{sign}(t)\, \kappa(v, t)\, c_{k-1}(u, v), \qquad \mathrm{sign}(t_{ak}) = -1,\ \mathrm{sign}(t_{kb}) = +1,$$

where the summand is understood to be 0 unless $v \lhd w$. (These are precisely the terms of (6.8) and (6.9) below.) The summand of t is therefore nonzero if and only if $v \lhd w$ and $u \overset{k-1}{\rightsquigarrow} v$, that is, σ is $(k-1)$-shaped and u-tight.

Lemma 6.15 (Classification of the Monk neighbors)**.** *Suppose the Monk summand of t is nonzero, and let $P = \mathrm{fix}_{k-1}(u, w)$. Then every nontrivial cycle of π not meeting t has exactly one element greater than k, and exactly one of the following holds.*

(U) *$t = t_{kB}$ with $B > k$; C contains exactly two elements greater than $k-1$, namely k and B; and $\mathrm{fix}_{k-1}(u, v) = P$. Writing $C = (a_p, \dots, a_1, B)$ with $a_j = k$, the cycles of σ are those of π other than C together with*

$$C' = (a_{j-1}, \dots, a_1, B) \quad \text{and} \quad C'' = (a_p, \dots, a_{j+1}, k).$$

(S) $t = t_{ak}$ *with* $a \in C$ *and* $B > k$*;* C *contains exactly two elements greater than* $k-1$*, namely* k *and* B*; in the notation of (U),* $a = a_m$ *for some* $m > j$*;* $\mathrm{fix}_{k-1}(u,v) = P$*; and the cycles of* σ *are those of* π *other than* C *together with*

$$C'_m = (a_p, \ldots, a_{m+1}, a_m, a_{j-1}, \ldots, a_1, B) \quad \text{and} \quad C''_m = (a_{m-1}, \ldots, a_{j+1}, k).$$

(P) $t = t_{ak}$ *with* $a \in C$ *and* $B = k$*; writing* $C = (a_p, \ldots, a_1, k)$ *we have* $a = a_p = \pi(k) = u^{-1}(w(k))$*;* $\mathrm{fix}_{k-1}(u,v) = P \sqcup \{w(k)\}$*; and the cycles of* σ *are those of* π *other than* C *together with* $C_- = (a_{p-1}, \ldots, a_1, k)$ *(trivial if* $p = 1$*).*

(M) $t = t_{ak}$ *with* $a \notin C$*; then* $B = k$*, the cycle* D *of* π *containing* a *(possibly* $D = \{a\}$*) satisfies* $D \subseteq [1, k-1]$*, and* $\mathrm{fix}_{k-1}(u,v) = P \setminus \{u(a)\}$*, which equals* P *unless* $D = \{a\}$*. The cycles of* σ *are those of* π *other than* C *and* D *together with the single cycle* E *with* $\mathrm{supp}(E) = \mathrm{supp}(C) \cup \mathrm{supp}(D)$ *obtained by setting* $E(a) = \pi(k)$*,* $E(k) = \pi(a)$ *and* $E = \pi$ *elsewhere.*

Proof. A cycle of π disjoint from $\mathrm{supp}(t)$ is a cycle of σ, hence has exactly one element greater than $k-1$, and since it does not contain k this element is greater than k. Recall that right multiplication of a permutation by a transposition t_{xy} merges the cycles of x and y if they are distinct, and otherwise splits their common cycle $(x_1, \ldots, x_n)$, $x = x_s$, $y = x_t$, $s < t$, into $(x_t, x_{s+1}, \ldots, x_{t-1})$ and $(x_s, x_{t+1}, \ldots, x_n, x_1, \ldots, x_{s-1})$.

Let $t = t_{kb}$. If $b \notin C$, the cycle of σ through k contains k and $b > k$, which is forbidden. So $b \in C$ and t splits C into a cycle through k and a cycle through b; each must contain exactly one element greater than $k-1$, so $C \cap (k-1, \infty) = \{k, b\}$ and $b = B$. The explicit form of the two cycles is the splitting formula applied to $(x_1, \ldots, x_{p+1}) = (a_p, \ldots, a_1, B)$ with $x_s = k$, $x_t = B$. Since v agrees with w at all positions less than k, $\mathrm{fix}_{k-1}(u,v) = P$. This is (U).

Let $t = t_{ak}$ with $a \in C$, so t splits C. If $B > k$, the piece not containing k must contain exactly one element greater than $k-1$, which forces $C \cap (k-1, \infty) = \{k, B\}$ and that piece to contain B. With $(x_1, \ldots, x_{p+1}) = (a_p, \ldots, a_1, B)$, $k = a_j$ and $a = a_m$: if $m < j$ the piece through a_m is $(a_m, a_{j-1}, \ldots, a_{m+1})$, which has no element greater than $k-1$ unless it is trivial ($m = j-1$), in which case the other piece $(k, a_{m-1}, \ldots, a_1, B, a_p, \ldots, a_{j+1})$ contains both k and B; both are forbidden. So $m > j$, and the splitting formula gives C'_m and C''_m. Here $v(a_m) = w(k) = u(a_{j-1}) \neq u(a_m)$ and $w(a_m) \neq u(a_m)$, so a_m is a fixed position neither for v nor for w, and $\mathrm{fix}_{k-1}(u,v) = P$. This is (S). If $B = k$, the piece not containing k has no element greater than $k-1$ and must be trivial, i.e. $\sigma(a) = a$, i.e. $\pi(k) = a$; in the notation $C = (a_p, \ldots, a_1, k)$ this is $a = a_p$, and $u(a_p) = u(\pi(k)) = w(k)$. Then σ has the cycle C_-, and $v(a_p) = w(k) = u(a_p)$ makes a_p a new fixed position with value $w(k) \notin P$. This is (P).

Finally let $t = t_{ak}$ with $a \notin C$, so t merges C with the cycle D of a into a cycle E containing k; E may contain no further element greater than $k-1$, so $C \subseteq [1, k]$, i.e. $B = k$, and $D \subseteq [1, k-1]$. We have $v(a) = w(k) = u(\pi(k)) \neq u(a)$ because $\pi(k) \in C \not\ni a$, so a is not a fixed position of v; it is one of w exactly when $D = \{a\}$. All other positions less than k agree. This is (M). □

The case $B > k$*: the up-neighbor and the split down-neighbors.*

Lemma 6.16 (Up/down pairing)**.** *Let* $\psi \in S_\infty$*, let* $k < B$*, let* $z_1, \ldots, z_s$ *(*$s \geq 0$*) be distinct positions less than* k*, let* $X = (z_s, \ldots, z_1, k, B)$ *and* $w = \psi X$*. Consider the following sequences of transpositions, each with product* X*:*

$$\begin{aligned} F_{\mathrm{std}} &: \ t_{kB},\, t_{z_1 B},\, \ldots,\, t_{z_s B} && \text{(the pivot factorization of } X\text{)}, \\ F_{\mathrm{up}} &: \ t_{z_1 k},\, \ldots,\, t_{z_s k},\, t_{kB}, && \\ F_m &: \ t_{z_1 k},\, \ldots,\, t_{z_{m-1} k},\, t_{z_m B},\, \ldots,\, t_{z_s B},\, t_{z_m k} && (1 \leq m \leq s). \end{aligned}$$

Call such a sequence a cover path *(from* ψ*) if each of its steps is a cover of the permutation reached before it, and define its* weight *to be the product of the* κ*'s of its steps.*

(a) *If* F_{std} *is a cover path, then* F_{up} *is a cover path of the same weight, and no* F_m *is a cover path.*

(b) *If* F_{up} *is a cover path but* F_{std} *is not, then exactly one* F_m *is a cover path, and its weight equals that of* F_{up}*.*

(c) *If some* F_m *is a cover path, then so is* F_{up}*.*

Proof. (The products are X: F_{up} is $(z_s, \ldots, z_1, k)\, t_{kB}$, and F_m is $(z_{m-1}, \ldots, z_1, k)\, (z_s, \ldots, z_m, B)\, t_{z_m k}$; both send $k \mapsto B \mapsto z_s \mapsto \cdots \mapsto z_1 \mapsto k$.) If $s = 0$ then $F_{\mathrm{std}} = F_{\mathrm{up}}$ and there is nothing to prove, so let $s \geq 1$.

Put

$$\beta = \psi(B), \qquad \zeta_0 = \psi(k), \qquad \zeta_i = \psi(z_i)\ (1 \le i \le s), \qquad I = \{\psi(r) \mid k < r < B\}.$$

The positions strictly between k and B are never moved, so I is the set of values found there at every stage of every path. Let $\phi_m = \psi t_{z_1 k} \cdots t_{z_{m-1} k}$ $(1 \le m \le s+1)$; thus $\phi_m(k) = \zeta_{m-1}$, $\phi_m(z_i) = \zeta_{i-1}$ for $i < m$, $\phi_m(z_i) = \zeta_i$ for $i \ge m$, $\phi_m(B) = \beta$, and ϕ_m agrees with ψ elsewhere. Let $J_i = \{\phi_i(r) \mid z_i < r < k\}$. We introduce the predicates

$$\begin{aligned} U_i &: \ (\zeta_i, \zeta_{i-1}) \text{ is } J_i\text{-admissible} && (1 \le i \le s), \\ T_i &: \ (\zeta_i, \beta) \text{ is } I\text{-admissible} && (0 \le i \le s), \\ A_i &: \ (\zeta_i, \zeta_{i-1}) \text{ is } (J_i \cup \{\beta\} \cup I)\text{-admissible} && (1 \le i \le s). \end{aligned}$$

U_i says that the i-th step of F_{up} (from ϕ_i) is a cover, and T_s that its last step is; so F_{up} is a cover path if and only if $U_1, \dots, U_s, T_s$ hold. For $1 \le m \le s+1$ let $\mathcal{S}_m$ be the statement that $t_{kB}, t_{z_m B}, \dots, t_{z_s B}$ is a cover path from ϕ_m; thus F_{std} is a cover path if and only if $\mathcal{S}_1$, and $\mathcal{S}_{s+1} = T_s$. The first step of $\mathcal{S}_m$ exchanges ζ_{m-1} at k with β at B across I; the step $t_{z_i B}$ $(i \ge m)$ exchanges ζ_i at z_i with ζ_{i-1} at B across the positions in (z_i, k), which carry the same values as in ϕ_i (the positions z_l, $l < i$, carry ζ_{l-1} and the others their ψ-values), the position k, which carries β, and the positions in (k, B). Hence

$$\mathcal{S}_m \iff T_{m-1} \wedge A_m \wedge \cdots \wedge A_s.$$

Let $\mathcal{D}_m$ $(1 \le m \le s)$ be the statement that $t_{z_m B}, \dots, t_{z_s B}, t_{z_m k}$ is a cover path from ϕ_m; then F_m is a cover path if and only if $U_1, \dots, U_{m-1}$ and $\mathcal{D}_m$ hold. The steps of $\mathcal{D}_m$ are: $D^{(m)}$, exchanging ζ_m at z_m with β at B across $J_m \cup \{\zeta_{m-1}\} \cup I$ (the position k now carries ζ_{m-1}); for $i > m$, $D^{(i)}$, exchanging ζ_i with ζ_{i-1} across $J_i^* \cup I$, where J_i^* consists of the values at the positions of $(z_i, k]$, so that $J_i^* = J_i \cup \{\beta\}$ if $z_m > z_i$ and $J_i^* = J_i \cup \{\zeta_{m-1}\}$ if $z_m < z_i$ (the position z_m carries β and k carries ζ_{m-1}); and D^{last}, exchanging β at z_m with ζ_{m-1} at k across $J^{**} = N_m \cup \{\zeta_{l-1} \mid l > m,\ z_l > z_m\}$, where N_m denotes the set of values carried by the positions of (z_m, k) other than $z_{m+1}, \dots, z_s$ (these values are the same in ϕ_m and at the end of $\mathcal{D}_m$), so that $J_m = N_m \cup \{\zeta_l \mid l > m,\ z_l > z_m\}$. For $i > m$ the set across which $D^{(i)}$ is taken contains J_i, and the sets of $D^{(m)}, \dots, D^{(s)}$ all contain ζ_{m-1}, the value at k.

Step 1: $\mathcal{S}_m \Rightarrow \mathcal{S}_{m+1}$, and $\mathcal{S}_m$ implies $U_m, \dots, U_s, T_s$. By Lemma 6.4(iii) applied with $(\alpha, \gamma, \beta) = (\zeta_m, \zeta_{m-1}, \beta)$ and $J = J_m$, the conjunction $T_{m-1} \wedge A_m$ implies $T_m \wedge U_m$. The first claim follows, and the second by iterating it down to $\mathcal{S}_{s+1} = T_s$, using $A_i \Rightarrow U_i$ (Lemma 6.4(i)). In particular $\mathcal{S}_1$ implies that F_{up} is a cover path.

Step 2: $\mathcal{D}_m$ implies $U_m, \dots, U_s, T_s$. For $i > m$, $D^{(i)}$ gives U_i by Lemma 6.4(i). By the same lemma, $D^{(m)}$ and $D^{(i)}$ $(i > m)$ give that (ζ_m, β) and (ζ_i, ζ_{i-1}) are I-admissible, so (ζ_s, β) is I-admissible by Lemma 6.4(ii): this is T_s. Likewise the pairs $(\zeta_m, \beta), (\zeta_{m+1}, \zeta_m), \dots, (\zeta_s, \zeta_{s-1})$ are $\{\zeta_{m-1}\}$-admissible, which as in the proof of Lemma 6.4(ii) means:

$$\begin{gathered} \text{the values } \beta, \zeta_m, \zeta_{m+1}, \dots, \zeta_s \text{ decrease on each side of } \zeta_{m-1}, \\ \text{with at most one jump from below } \zeta_{m-1} \text{ to above it.} \end{gathered} \tag{6.4}$$

For U_m, admissibility of (ζ_m, β) with respect to $\{\zeta_{m-1}\}$ leaves three possibilities: (1) $\zeta_{m-1} > \beta > \zeta_m$; (2) $\zeta_m > \zeta_{m-1} > \beta$; (3) $\beta > \zeta_m > \zeta_{m-1}$. We use $D^{(m)}$ restricted to J_m and D^{last} restricted to N_m, and treat the remaining elements ζ_l $(l > m,\ z_l > z_m)$ of J_m by (6.4). In case (1), $J_m \cap (\zeta_m, \beta) = \emptyset$ and $N_m \cap (\beta, \zeta_{m-1}) = \emptyset$, while by (6.4) each ζ_l $(l > m)$ is either less than ζ_m or greater than ζ_{m-1}; hence $J_m \cap (\zeta_m, \zeta_{m-1}) = \emptyset$. In case (2), $J_m \subseteq (\beta, \zeta_m)$ and $N_m \cap (\beta, \zeta_{m-1}) = \emptyset$ give $N_m \subseteq (\zeta_{m-1}, \zeta_m)$, while by (6.4) the jump occurs at (β, ζ_m) and so $\zeta_{m-1} < \zeta_l < \zeta_m$ for $l > m$; hence $J_m \subseteq (\zeta_{m-1}, \zeta_m)$. In case (3), $J_m \cap (\zeta_m, \beta) = \emptyset$ and $N_m \subseteq (\zeta_{m-1}, \beta)$ give $N_m \subseteq (\zeta_{m-1}, \zeta_m)$, while by (6.4) all of $\beta, \zeta_m, \dots, \zeta_s$ lie above ζ_{m-1} and decrease, so again $\zeta_{m-1} < \zeta_l < \zeta_m$ for $l > m$; hence $J_m \subseteq (\zeta_{m-1}, \zeta_m)$. In each case this is U_m. Together with the prefix $U_1, \dots, U_{m-1}$ this proves (c).

For the remainder assume that F_{up} is a cover path. Then the cycle $(z_s, \dots, z_1, k)$ is ψ-tight, so by Corollary 6.13 the cyclic sequence $\Gamma = (\zeta_0, \zeta_1, \dots, \zeta_s)$ is unimodal; let $\delta \in \{0, \dots, s\}$ be the index of its ascent ($\zeta_\delta > \zeta_{\delta-1}$ if $\delta \ge 1$; $\delta = 0$ means the ascent is the wrap (ζ_s, ζ_0)), and let $g \in \{0, \dots, s\}$ be the insertion gap of β in Γ, where gap $g \ge 1$ is the gap (ζ_{g-1}, ζ_g) and gap 0 is the wrap. Explicitly: if $\min \Gamma < \beta < \max \Gamma$, then g is determined by $\zeta_{g-1} > \beta > \zeta_g$ (indices modulo $s+1$); otherwise $g = \delta$.

Step 3: $\mathcal{S}_m$ implies $m > g$. $\mathcal{S}_m$ says that the cycle $(z_s, \dots, z_m, k, B)$ is ϕ_m-tight, so by Corollary 6.13 the cyclic sequence $(\beta, \zeta_{m-1}, \dots, \zeta_s)$ is unimodal; that is, the wrap gap (ζ_s, ζ_{m-1}) of the block $\Gamma' = (\zeta_{m-1}, \dots, \zeta_s)$

is the insertion gap of β in Γ'. Suppose $1 \le m \le g$. Then ζ_{g-1}, ζ_g both lie in Γ' and are consecutive there. If $\min\Gamma < \beta < \max\Gamma$ then $\zeta_{g-1} > \beta > \zeta_g$, so the insertion gap of β in Γ' is (ζ_{g-1}, ζ_g), which is not the wrap gap of Γ' since $g-1 \neq s$. If β lies outside $[\min\Gamma, \max\Gamma]$ then $g = \delta$, so $\zeta_g = \max\Gamma$ and $\zeta_{g-1} = \min\Gamma$ lie in Γ', whence β lies outside the range of Γ' and its insertion gap in Γ' is the ascent gap (ζ_{g-1}, ζ_g) of Γ', again not the wrap gap. In both cases $\mathcal{S}_m$ fails.

Step 4: $\mathcal{S}_{g+1}$ holds; hence $\mathcal{S}_m$ holds if and only if $m > g$. We must verify T_g and A_i for $i > g$; by Lemma 6.4(i) we treat the sets J_i, $\{\beta\}$ and I separately. Admissibility of (ζ_i, ζ_{i-1}) with respect to J_i is U_i. With respect to $\{\beta\}$: if $i > g$ and $i \neq \delta$, the gap (ζ_{i-1}, ζ_i) is a descent gap and $\beta \notin (\zeta_i, \zeta_{i-1})$, for otherwise it would be the insertion gap of β; if $i = \delta > g$, then $\beta \in (\min\Gamma, \max\Gamma) = (\zeta_{\delta-1}, \zeta_\delta)$, for otherwise $g = \delta$. With respect to I, and for T_g: the sequence $s_0 = \beta, s_1 = \zeta_g, \dots, s_n = \zeta_s$ is a consecutive block of the unimodal sequence obtained by inserting β into Γ, hence unimodal, and T_s says that (s_n, s_0) is I-admissible. If $s_0 > s_1 > \cdots > s_n$, then T_s is classical, $I \cap (s_n, s_0) = \emptyset$, and every pair (s_i, s_{i-1}) is I-admissible. Otherwise the linear ascent is at some $j \ge 1$, so $s_j > \cdots > s_n > s_0 > \cdots > s_{j-1}$; then T_s is quantum, $I \subseteq (s_0, s_n)$, and the pairs (s_i, s_{i-1}) with $i < j$ lie below s_0, the pair (s_j, s_{j-1}) has $I \subseteq (s_0, s_n) \subseteq (s_{j-1}, s_j)$, and the pairs with $i > j$ lie above s_n; all are I-admissible. The pairs (s_i, s_{i-1}) are exactly (ζ_g, β) and (ζ_i, ζ_{i-1}), $i > g$. This proves $\mathcal{S}_{g+1}$, and the second assertion follows from Steps 1 and 3.

Step 5: $\mathcal{D}_m$ implies $m = g$. Restricting $D^{(m)}$ to $\{\zeta_{m-1}\}$ leaves the three cases (1)–(3) of Step 2. In case (1) the gap (ζ_{m-1}, ζ_m) is a descent gap containing β, so $g = m$. In case (2) the gap is the ascent gap, $m = \delta$, and $\beta < \zeta_{m-1} = \min\Gamma$, so $g = \delta = m$. In case (3), $m = \delta$ and $\beta > \zeta_m = \max\Gamma$, so again $g = m$.

Step 6: if $g \ge 1$ then $\mathcal{D}_g$ holds. Write $a = z_g$. We are in one of the cases (1) $\zeta_{g-1} > \beta > \zeta_g$ ($g \neq \delta$), (2) $g = \delta$ and $\beta < \zeta_{g-1} = \min\Gamma$, (3) $g = \delta$ and $\beta > \zeta_g = \max\Gamma$. For $D^{(g)}$: admissibility of (ζ_g, β) with respect to I is T_g (Step 4); with respect to $\{\zeta_{g-1}\}$ it holds in each of the three cases by inspection; with respect to J_g it follows from U_g: in case (1), $J_g \cap (\zeta_g, \zeta_{g-1}) = \emptyset$ gives $J_g \cap (\zeta_g, \beta) = \emptyset$; in case (2), $J_g \subseteq (\zeta_{g-1}, \zeta_g) \subseteq (\beta, \zeta_g)$; in case (3), $J_g \subseteq (\zeta_{g-1}, \zeta_g)$ is disjoint from (ζ_g, β). For $D^{(i)}$, $i > g$: if $z_g > z_i$ the required statement is A_i (Step 4); if $z_g < z_i$ then $z_g \in (z_i, k)$, so $\zeta_{g-1} = \phi_i(z_g) \in J_i$ already, $J_i^* = J_i$, and the statement follows from U_i and the I-admissibility established in Step 4. For D^{last}, the pair (β, ζ_{g-1}) must be admissible with respect to $N_g \subseteq J_g$ and to the values ζ_{l-1}, $l > g$, $z_l > z_g$: in case (1) it is classical, and $(\beta, \zeta_{g-1}) \subseteq (\zeta_g, \zeta_{g-1})$ contains no element of J_g (by U_g) and no ζ_l (as ζ_{g-1}, ζ_g are consecutive in the decreasing order of Γ); in case (2) it is classical, and (β, ζ_{g-1}) lies below $\min\Gamma$ and below $J_g \subseteq (\zeta_{g-1}, \zeta_g)$; in case (3) it is quantum, and $J_g \subseteq (\zeta_{g-1}, \zeta_g) \subseteq (\zeta_{g-1}, \beta)$ while every ζ_{l-1} with $l > g$ lies in $(\zeta_{g-1}, \zeta_g] \subseteq (\zeta_{g-1}, \beta)$.

Step 7: weights. Suppose F_{up} is a cover path and $g \ge 1$. The two paths F_{up} and F_g share the prefix $t_{z_1 k}, \dots, t_{z_{g-1} k}$; after it, the quantum steps of F_{up} are $t_{z_\delta k}$ (if $\delta \ge g$) and t_{kB} (if $\zeta_s > \beta$), while those of F_g are $t_{z_g B}$ (if $\zeta_g > \beta$), $t_{z_\delta B}$ (if $\delta > g$) and $t_{z_g k}$ (if $\beta > \zeta_{g-1}$). In case (1) with $\delta > g$ we have $\zeta_s > \zeta_0 \ge \zeta_{g-1} > \beta$, and both weights are $q_{z_\delta k} q_{kB} = q_{z_\delta B}$; in case (1) with $\delta < g$ we have $\zeta_s < \zeta_g < \beta$ and both weights are 1; in case (2) both are $q_{z_g k} q_{kB} = q_{z_g B}$; in case (3) both are $q_{z_g k}$. If instead $g = 0$, so that $\mathcal{S}_1$ holds, the quantum steps of F_{std} are t_{kB} (if $\zeta_0 > \beta$) and $t_{z_\delta B}$ (if $\delta \ge 1$); if $\zeta_s > \beta > \zeta_0$ (so $\delta \ge 1$) both F_{std} and F_{up} have weight $q_{z_\delta B}$; if $\beta < \min\Gamma$ (so $\delta = 0$) both have weight q_{kB}; and if $\beta > \max\Gamma$ both have weight 1.

Conclusion. (a): $\mathcal{S}_1$ gives F_{up} (Step 1) with the same weight (Step 7), and $g = 0$ (Step 3), so no $\mathcal{D}_m$ holds (Step 5). (b): if F_{up} is a cover path and $\mathcal{S}_1$ fails, then $g \ge 1$ (Step 4), $\mathcal{D}_g$ holds (Step 6) and no other $\mathcal{D}_m$ does (Step 5); as $U_1, \dots, U_{g-1}$ hold, F_g is the unique F_m which is a cover path, and its weight is that of F_{up} (Step 7). (c) is Step 2. □

Lemma 6.17. *Suppose $u, w \in S_\infty$ and let $k > 1$ be an integer. If $u \overset{k}{\rightsquigarrow} w$ but $u \not\overset{k-1}{\rightsquigarrow} w$, then there is a unique $b > k$ such that $wt_{kb} \lhd w$ and $u \overset{k-1}{\rightsquigarrow} wt_{kb}$, namely $b = B$. Furthermore, in that case there does not exist an $a < k$ such that $wt_{ak} \lhd w$ and $u \overset{k-1}{\rightsquigarrow} wt_{ak}$, and*

$$\kappa(wt_{kB}, t_{kB})\, q_{k-1}(u, wt_{kB}) = q_k(u, w), \qquad \mathrm{fix}_{k-1}(u, wt_{kB}) = \mathrm{fix}_k(u, w).$$

Proof. By Lemma 6.8, π is k-shaped and u-tight but not $(k-1)$-shaped, so $u(k) \neq w(k)$ and $B > k$ (Lemma 6.14). Write $C = (a_p, \dots, a_1, B)$ with $a_j = k$, let ρ be the product of the other nontrivial cycles of π, let $C' = (a_{j-1}, \dots, a_1, B)$, and put $\psi = u\rho C'$ and $X = (a_p, \dots, a_{j+1}, k, B)$, so that $w = \psi X$ and X has the form of Lemma 6.16 with $s = p - j$ and $z_i = a_{j+i}$. Take the pivot chain of π consisting of the pivot factorization of ρ, then that of C; by Lemma 6.7(iii),(v) all its steps are covers, ρ is u-tight, C' is $u\rho$-tight (it is C truncated after $j-1$ factors), and F_{std} is a cover path from ψ. By Lemma 6.16(a), F_{up} is a cover path of the same weight. Its first s steps form the pivot factorization of $C'' = (a_p, \dots, a_{j+1}, k)$, so $\rho C' C'' = u^{-1} wt_{kB}$ is u-tight

by Lemma 6.7(iii),(iv); it is $(k-1)$-shaped, its cycles being those of ρ (each with exactly one element greater than k), C' (top B, other elements less than k) and C'' (top k, other elements less than k). Hence $u \overset{k-1}{\rightsquigarrow} wt_{kB}$, and the last step of F_{up} says $wt_{kB} \lhd w$. Uniqueness of b is Lemma 6.15(U). The weights are $\mathbf{q}_u(\rho)\mathbf{q}_u(C')$ times the weights of F_{std}, respectively F_{up}, which agree; and $\mathrm{fix}_{k-1}(u, wt_{kB}) = \mathrm{fix}_{k-1}(u,w) = \mathrm{fix}_k(u,w)$ since $u(k) \neq w(k)$.

Suppose finally that $a < k$ satisfies $wt_{ak} \lhd w$ and $u \overset{k-1}{\rightsquigarrow} wt_{ak}$. By Lemma 6.15 we are in case (S): $a = a_m$ with $m > j$, and $\sigma = \rho C'_m C''_m$ is u-tight. Taking the pivot chain of σ in the order: factors of ρ, the first $j-1$ factors of C'_m (which are those of C'), the factors of C''_m, the remaining factors $t_{a_m B}, \dots, t_{a_p B}$ of C'_m, and appending the cover $t_{a_m k}$, we obtain that F_{m-j} is a cover path from ψ, contradicting Lemma 6.16(a). □

The case $B = k$: the peel and the merges.

Lemma 6.18 (Peel)**.** *Let $\omega \in S_\infty$ and let $C = (a_p, \dots, a_1, k)$ $(p \geq 1)$ be a cycle with top k; put $w = \omega C$, $C_- = (a_{p-1}, \dots, a_1, k)$ and $v = wt_{a_p k} = \omega C_-$. Then C is ω-tight if and only if C_- is ω-tight and $v \lhd w$, and in that case $\kappa(v, t_{a_p k})\, \mathbf{q}_\omega(C_-) = \mathbf{q}_\omega(C)$.*

Proof. The pivot factorization of C is that of C_- followed by $t_{a_p k}$, and $\omega C_- = v$. Apply Lemma 6.7(iii). □

Lemma 6.19 (Merge)**.** *Let $\omega \in S_\infty$, $k > 1$, $\eta = \omega(k)$, and let D be a cycle with* $\mathrm{supp}(D) \subseteq [1, k-1]$*, allowing D to be trivial, $D = \{a\}$ with $a \leq k-1$. Put $w = \omega D$, so $w(k) = \eta$. For $a \in$* $\mathrm{supp}(D)$ *let $\tau_a = D t_{ak}$, the cycle with support* $\mathrm{supp}(D) \cup \{k\}$ *sending $a \mapsto k \mapsto D(a)$, and $v_a = wt_{ak} = \omega\tau_a$. Then:*

(a) *there is at most one $a \in$* $\mathrm{supp}(D)$ *such that τ_a is ω-tight and $v_a \lhd w$;*
(b) *such an a exists if and only if D is ω-tight and* $\max \mathrm{supp}(D) = k-1$ *(for trivial $D = \{a\}$: if and only if $a = k-1$);*
(c) *for this a, $\kappa(v_a, t_{ak})\, \mathbf{q}_\omega(\tau_a) = q_{k-1}\, \mathbf{q}_\omega(D)$.*

Proof. Trivial $D = \{a\}$. Then $w = \omega$, $\tau_a = t_{ak}$, and τ_a is ω-tight if and only if $w \lhd wt_{ak}$. If also $wt_{ak} \lhd w$, then both $\ell(wt_{ak}) - \ell(w)$ and its negative lie in $\{1, 1-2(k-a)\}$, which forces $k - a = 1$. Conversely for $a = k-1$ the transposition $t_{k-1,k}$ changes the length by ± 1, so both relations hold, one classical and one quantum with $\kappa = q_{k-1}$; thus $\kappa(v_a, t_{ak})\mathbf{q}_\omega(\tau_a) = q_{k-1} = q_{k-1}\mathbf{q}_\omega(D)$.

Nontrivial $D = (d_r, \dots, d_1, d)$, $r \geq 1$, $d = \max \mathrm{supp}(D)$. Put $d_0 = d$, $\gamma_0 = \omega(d)$, $\gamma_i = \omega(d_i)$ and $\epsilon = k - d \geq 1$. For $a = d_m$ $(0 \leq m \leq r)$ the cycle $\tau_m := \tau_{d_m}$ is

$$\tau_m = (d_{m-1}, \dots, d_1, d, d_r, \dots, d_m, k),$$

whose pivot factorization

$$t_{d_m k}\, t_{d_{m+1} k} \cdots t_{d_r k}\, t_{dk}\, t_{d_1 k} \cdots t_{d_{m-1} k}$$

exchanges, in order, the values $\eta, \gamma_m, \gamma_{m+1}, \dots, \gamma_r, \gamma_0, \gamma_1, \dots, \gamma_{m-1}$: the sequence of values of τ_m in the sense of Corollary 6.13 is the cyclic sequence $\Gamma = (\gamma_0, \dots, \gamma_r)$ of D with η inserted into the gap (γ_{m-1}, γ_m) (the wrap gap (γ_r, γ_0) when $m = 0$). Note $w(d_m) = \gamma_{m-1}$ for $m \geq 1$ and $w(d) = \gamma_r$; we write $w(a)$ for the value at $a = d_m$ in either case.

Necessity. Suppose τ_m is ω-tight and $v_m := v_{d_m} \lhd w$. By Corollary 6.13 the cyclic sequence of τ_m is unimodal; deleting η shows that Γ is unimodal, and then m must be the insertion gap of η in Γ. This already proves (a). Let δ be the ascent index of Γ ($\gamma_\delta > \gamma_{\delta-1}$, or $\delta = 0$ for the wrap), so that by Corollary 6.13 and Definition 6.6

$$L_\omega(D) = r - 2(d - d_\delta), \qquad \mathbf{q}_\omega(D) = q_{d_\delta d}$$

(read as r and 1 when $\delta = 0$). Now compare lengths. Tightness gives $\ell(v_m) - \ell(\omega) = L_\omega(\tau_m)$, Lemma 6.7(ii) gives $\ell(w) - \ell(\omega) \geq L_\omega(D)$ with equality if and only if D is ω-tight, and $v_m \lhd w$ gives $\ell(v_m) - \ell(w) = -1$ if $w(a) > \eta$ and $\ell(v_m) - \ell(w) = 2(k-a) - 1$ if $w(a) < \eta$. Hence

$$L_\omega(\tau_m) - L_\omega(D) \;\geq\; \begin{cases} -1 & \text{if } w(a) > \eta, \\ 2(k-a) - 1 & \text{if } w(a) < \eta, \end{cases} \tag{6.5}$$

with equality if and only if D is ω-tight. We compute the left side from Corollary 6.13 in the three possible configurations of the insertion gap.

(I) $\min \Gamma < \eta < \max \Gamma$, so $\gamma_{m-1} > \eta > \gamma_m$ and $m \neq \delta$. The ascent of the sequence of τ_m is the (interior) step onto γ_δ, whose position is d_δ, so $L_\omega(\tau_m) = (r+1) - 2(k - d_\delta)$ and $L_\omega(\tau_m) - L_\omega(D) = 1 - 2\epsilon$. Here $w(a) = \gamma_{m-1} > \eta$, so (6.5) reads $1 - 2\epsilon \geq -1$, forcing $\epsilon = 1$ and equality.

(II) $\eta < \min\Gamma$, so $m = \delta$. The ascent of the sequence of τ_m is its first step (η, γ_δ), at position d_δ; again $L_\omega(\tau_m) - L_\omega(D) = 1 - 2\epsilon$, and $w(a) = \gamma_{\delta-1} > \eta$ forces $\epsilon = 1$ and equality.

(III) $\eta > \max\Gamma$, so $m = \delta$. The sequence of τ_m is decreasing, its ascent being the wrap, so $L_\omega(\tau_m) = r+1$ and $L_\omega(\tau_m) - L_\omega(D) = 1+2(d-d_\delta)$. Here $w(a) = \gamma_{\delta-1} < \eta$, and (6.5) reads $1+2(d-d_\delta) \geq 2(k-d_\delta)-1$, i.e. $\epsilon \leq 1$; so $\epsilon = 1$ and equality.

In every case $d = k-1$ and D is ω-tight, proving the "only if" part of (b).

Sufficiency. Suppose D is ω-tight and $d = k-1$; then Γ is unimodal (Corollary 6.13). Let m be the insertion gap of η and $a = d_m$. The computations above show that (6.5) holds with equality, and since $\ell(w) - \ell(\omega) = L_\omega(D)$, the tightness of τ_m is equivalent to $\ell(v_m) - \ell(w)$ taking the value -1 (if $w(a) > \eta$) or $2(k-a)-1$ (if $w(a) < \eta$), i.e. to $v_m \lhd w$. It therefore suffices to show $wt_{ak} \lhd w$, that is, that $(w(a), \eta)$ is J-admissible for $J = \{w(r) \mid a < r < k\}$. The positions in (a,k) are: $d = k-1$ (if $a \neq d$), carrying γ_r; the $d_l > a$, carrying γ_{l-1}; and positions outside $\mathrm{supp}(D)$, carrying their ω-values. If $a = d = k-1$ the set J is empty and we are done; so assume $m \geq 1$.

(I) $\gamma_{m-1} > \eta > \gamma_m$: the pair $(w(a), \eta) = (\gamma_{m-1}, \eta)$ is classical and we need $J \cap (\eta, \gamma_{m-1}) = \emptyset$. No γ_l lies in (γ_m, γ_{m-1}), these being consecutive in the decreasing order of Γ. The m-th step of the pivot chain of D is classical, so by (6.2) every position $r \in (d_m, d)$ outside $\mathrm{supp}(D)$ has $\omega(r) \notin (\gamma_m, \gamma_{m-1})$.

(II) $m = \delta$, $\eta < \gamma_{\delta-1} = \min\Gamma$: the pair $(\gamma_{\delta-1}, \eta)$ is classical and we need $J \cap (\eta, \gamma_{\delta-1}) = \emptyset$; all $\gamma_l \geq \gamma_{\delta-1}$, and by (6.3) for the quantum step δ of D the positions $r \in (d_\delta, d)$ outside $\mathrm{supp}(D)$ have $\omega(r) \in (\gamma_{\delta-1}, \gamma_\delta)$.

(III) $m = \delta$, $\eta > \gamma_\delta = \max\Gamma$: the pair $(\gamma_{\delta-1}, \eta)$ is quantum and we need $J \subseteq (\gamma_{\delta-1}, \eta)$; the values γ_l occurring in J are different from $\gamma_{\delta-1}$ (which sits at a) and lie in $(\gamma_{\delta-1}, \gamma_\delta]$, and the remaining positions have $\omega(r) \in (\gamma_{\delta-1}, \gamma_\delta)$ by (6.3).

This proves the "if" part of (b).

Weights. In cases (I) and (II), $\kappa(v_a, t_{ak}) = 1$ and $\mathbf{q}_\omega(\tau_m) = q_{d_\delta k} = q_{d_\delta, k-1} q_{k-1} = \mathbf{q}_\omega(D) q_{k-1}$ (read $q_{dk} = q_{k-1}$ and $\mathbf{q}_\omega(D) = 1$ when $\delta = 0$). In case (III), $\kappa(v_a, t_{ak}) = q_{ak} = q_{d_\delta k}$ and $\mathbf{q}_\omega(\tau_m) = 1$, giving the same product. This is (c). □

The cancellations. We now assemble the Monk summands. Throughout, ρ denotes the product of the nontrivial cycles of π other than C and other than the cycle D of π containing $k-1$ (we set $D = \{k-1\}$ if $k-1$ is fixed by π, and we do not single out D when $k-1 \in C$). By Lemma 6.15, Lemma 6.8 and Lemma 6.7(iv), each of the quantities $c_{k-1}(u,w)$, $c_{k-2}(u,w)$, $c_k(u,w)$ and each Monk summand vanishes unless ρ is u-tight and each cycle of ρ contains exactly one element greater than k; indeed each requires ρ to be a tight product of cycles with exactly one element greater than k or than $k-1$ or than $k-2$, and these conditions agree for cycles containing neither k nor $k-1$. (For $c_{k-2}(u,w)$, note that $k-1$ lies in C or in D, hence not in a cycle of ρ.) *We assume this from now on*; otherwise all terms considered below are zero.

Lemma 6.20 (Reflection–drop cancellation)**.** *Suppose $u, w \in S_\infty$, let $k > 1$, and suppose $u(k) = w(k)$. Then*

$$\sum_{a<k,\ wt_{ak} \lhd w} \mathrm{sign}(a-k)\, \kappa_{ak}\, c_{k-1}(u, wt_{ak}) \ + \sum_{b>k,\ wt_{kb} \lhd w} \mathrm{sign}(b-k)\, \kappa_{kb}\, c_{k-1}(u, wt_{kb}) \ + \ q_{k-1}\, c_{k-2}(u,w) = 0,$$

where $\kappa_{rs} = \kappa(wt_{rs}, t_{rs})$. In fact the two reflection sums contribute at most one nonzero summand in total: it occurs precisely when $u \overset{k-2}{\rightsquigarrow} w$ (equivalently, when the quantum term is nonzero), it is indexed by some $a < k$, and it equals $-q_{k-1} c_{k-2}(u,w)$.

Proof. Here $C = \{k\}$, so by Lemma 6.15 only summands of type (M) can be nonzero: $t = t_{ak}$ with the π-cycle D_a of a contained in $[1, k-1]$. If D_a is nontrivial then it is not a cycle of ρ (those have an element greater than k), so $D_a = D$ and $\max D = k-1$; otherwise a is a fixed point of π. Let a be such a candidate and put $\omega_a = wD_a^{-1}$, so that $w = \omega_a D_a$, $\omega_a(k) = w(k)$, and $\omega_a = u\rho$ when $D_a = D$. Then $u^{-1} wt_{ak} = (\pi D_a^{-1})\, \tau_a$ with $\tau_a = D_a t_{ak}$ as in Lemma 6.19, where πD_a^{-1} is the product of the cycles of π other than D_a; by Lemma 6.7(iv), $u^{-1} wt_{ak}$ is u-tight if and only if πD_a^{-1} is u-tight and τ_a is ω_a-tight. Now apply Lemma 6.19 to (ω_a, D_a). If a is a fixed point, the summand can be nonzero only for $a = k-1$, which requires $k-1$ to be fixed, i.e. $D = \{k-1\}$; in that case $\pi D_a^{-1} = \rho$ and the summand is nonzero. If $D_a = D$ is nontrivial, at most one $a \in D$ works, and one does if and only if D is $u\rho$-tight. Altogether at most one a gives a nonzero summand, and one exists if and only if either $D = \{k-1\}$, or D is nontrivial with $\max D = k-1$ and

$u\rho$-tight. On the other hand $u \overset{k-2}{\rightsquigarrow} w$ if and only if $\pi = \rho D$ is $(k-2)$-shaped and u-tight, i.e. (given our standing assumption on ρ) if and only if D is trivial or has $k-1$ as its unique element greater than $k-2$, and D is $u\rho$-tight: the same condition. When it holds, with $v = wt_{ak}$ the unique contributing neighbor and $\omega = \omega_a = u\rho$, Lemma 6.19(c) and Lemma 6.8 give

$$\kappa_{ak}\, q_{k-1}(u,v) = \mathbf{q}_u(\rho)\, \kappa_{ak} \mathbf{q}_\omega(\tau_a) = \mathbf{q}_u(\rho)\, q_{k-1} \mathbf{q}_\omega(D) = q_{k-1}\, q_{k-2}(u,w),$$

and Lemma 6.15(M) gives $\mathrm{fix}_{k-1}(u,v) = \mathrm{fix}_{k-1}(u,w) \setminus \{u(a)\}$, which equals $\mathrm{fix}_{k-2}(u,w)$: if $D = \{k-1\}$ then $a = k-1$ and we remove the fixed value $u(k-1)$; if D is nontrivial then neither a nor $k-1$ is a fixed position of w. Hence the summand is $-\kappa_{ak} c_{k-1}(u,v) = -q_{k-1} c_{k-2}(u,w)$, cancelling the quantum term. If the condition fails, both the reflection sums and the quantum term vanish. □

Lemma 6.21 (Moved reflection cancellation)**.** *Let $u, w \in S_\infty$, let $k > 1$, and suppose $u(k) \neq w(k)$, so that C is nontrivial.*

(a) *If $B > k$, then $c_{k-1}(u,w) = c_{k-2}(u,w) = 0$, and the sum of the Monk summands equals $c_k(u,w)$. More precisely: if $u \overset{k}{\rightsquigarrow} w$ the only nonzero Monk summand is that of t_{kB}, and it equals $c_k(u,w)$; if $u \overset{k}{\not\rightsquigarrow} w$, the nonzero Monk summands are either none or exactly two, one of type (U) and one of type (S), with equal absolute value and opposite signs.*

(b) *If $B = k$, then $c_k(u,w) = 0$, no summand of type (U) occurs, and*

$$(y_{w(k)} - z_1)\, c_{k-1}(u,w) \;-\; \sum_{a<k,\ wt_{ak} \lhd w} \kappa_{ak}\, c_{k-1}(u, wt_{ak}) \;+\; q_{k-1} c_{k-2}(u,w) = 0.$$

In fact the diagonal term cancels against the summand of type (P), and the quantum term against the (at most one) summand of type (M).

Proof. (a) C contains k and B, two elements greater than $k-1$, so π is neither $(k-1)$- nor $(k-2)$-shaped and $c_{k-1}(u,w) = c_{k-2}(u,w) = 0$. By Lemma 6.15, a nonzero Monk summand is of type (U) or (S), and requires $C \cap (k-1, \infty) = \{k, B\}$; together with the standing assumption on ρ this means that π is k-shaped, which we may assume (otherwise all Monk summands vanish and $c_k(u,w) = 0$). Write $C = (a_p, \ldots, a_1, B)$ with $a_j = k$, $C' = (a_{j-1}, \ldots, a_1, B)$, $\psi = u\rho C'$ and $X = (a_p, \ldots, a_{j+1}, k, B)$, so that $w = \psi X$ as in Lemma 6.16 with $s = p - j$, $z_i = a_{j+i}$. If C' is not $u\rho$-tight then nothing is tight (Lemma 6.7(iv),(v): C' is a truncation of C and of each C'_m, and a cycle of $u^{-1}wt_{kB}$), all Monk summands vanish, and $c_k(u,w) = 0$. So assume C' is $u\rho$-tight. Then, exactly as in the proof of Lemma 6.17: $u \overset{k}{\rightsquigarrow} w$ if and only if F_{std} is a cover path from ψ; the summand of type (U) is nonzero if and only if F_{up} is a cover path, and then equals $\mathbf{q}_u(\rho)\mathbf{q}_u(C')\,\mathrm{wt}(F_{\mathrm{up}})\,\mathbf{E}(y_P; z)$ with $P = \mathrm{fix}_{k-1}(u,w)$; and the summand of type (S) indexed by a_m $(m > j)$ is nonzero if and only if F_{m-j} is a cover path, and then equals $-\mathbf{q}_u(\rho)\mathbf{q}_u(C')\,\mathrm{wt}(F_{m-j})\,\mathbf{E}(y_P; z)$ (Lemma 6.15(S) for the fixed set). If $u \overset{k}{\rightsquigarrow} w$, Lemma 6.16(a) and Lemma 6.17 show that the only nonzero summand is that of t_{kB} and that it equals $q_k(u,w)\mathbf{E}(y_{\mathrm{fix}_k(u,w)}; z) = c_k(u,w)$. If $u \overset{k}{\not\rightsquigarrow} w$, Lemma 6.16(b),(c) show that either no summand is nonzero, or exactly two are, namely (U) and one (S), with equal weights; they cancel, and $c_k(u,w) = 0$.

(b) $C \subseteq [1,k]$ has no element greater than k, so π is not k-shaped and $c_k(u,w) = 0$; type (U) is excluded by Lemma 6.15. Write $C = (a_p, \ldots, a_1, k)$. For the diagonal and peel terms, let $\omega_0 = u\rho D$ if $k-1 \notin C$ and $\omega_0 = u\rho$ if $k-1 \in C$ (where D is the cycle of $k-1$, trivial or not), so that $w = \omega_0 C$ and ω_0 agrees with u on $\mathrm{supp}(C)$. The diagonal term is nonzero if and only if π is $(k-1)$-shaped and tight, i.e. (given the standing assumption) D is trivial or has an element greater than k, D is $u\rho$-tight, and C is ω_0-tight (the conditions on D being vacuous when $k-1 \in C$). The summand of type (P) is nonzero if and only if $\rho D C_-$ is $(k-1)$-shaped and tight and $v = wt_{a_p k} \lhd w$, i.e. the same conditions on D together with: C_- is ω_0-tight and $v \lhd w$. By Lemma 6.18 these two sets of conditions are equivalent, and when they hold, $\kappa_{a_p k} q_{k-1}(u,v) = \mathbf{q}_u(\rho D)\kappa_{a_p k}\mathbf{q}_{\omega_0}(C_-) = \mathbf{q}_u(\rho D)\mathbf{q}_{\omega_0}(C) = q_{k-1}(u,w)$, while $\mathrm{fix}_{k-1}(u,v) = \mathrm{fix}_{k-1}(u,w) \sqcup \{w(k)\}$ by Lemma 6.15(P). Since $(y_{w(k)} - z_1)\mathbf{E}(y_{\mathrm{fix}_{k-1}(u,w)}; z) = \mathbf{E}(y_{\mathrm{fix}_{k-1}(u,w) \cup \{w(k)\}}; z)$, the diagonal term and the (negatively signed) summand of type (P) cancel.

For the merge and quantum terms, a summand of type (M) requires $a \notin C$ with the cycle D_a of a contained in $[1, k-1]$; as in the proof of Lemma 6.20, either a is fixed by π or $D_a = D$ with $\max D = k-1$ (in particular $k-1 \notin C$). If $k-1 \in C$, then $k-1$ is not fixed and no such D exists; since a fixed-point candidate must be $a = k-1$ by the trivial case of Lemma 6.19 (applied as below), there are no nonzero

summands of type (M), and $c_{k-2}(u,w)=0$ since C contains the two elements $k-1,k$ greater than $k-2$. So suppose $k-1\notin C$. For a candidate a put $\omega_a = wD_a^{-1}$, so that $w=\omega_a D_a$ and $\omega_a(k)=w(k)$; then $u^{-1}wt_{ak}=\rho_a E$, where ρ_a is the product of the cycles of π other than C and D_a and $E=C\tau_a$ is the cycle of Lemma 6.15(M), $\tau_a = D_a t_{ak}$. The pivot factorization of E is that of C followed by that of τ_a, so by Lemma 6.7(iii),(iv), $u^{-1}wt_{ak}$ is u-tight if and only if ρ_a is u-tight, C is $u\rho_a$-tight and τ_a is ω_a-tight (note $u\rho_a C=\omega_a$), and then $\mathbf{q}_u(E)=\mathbf{q}_u(C)\mathbf{q}_{\omega_a}(\tau_a)$. By Lemma 6.19 applied to (ω_a,D_a): a fixed-point candidate contributes only if $a=k-1$, i.e. $D=\{k-1\}$; and if $D_a=D$ is nontrivial, at most one $a\in D$ contributes, and one does if and only if D is ω_a-tight, where now $\omega_a = u\rho C =: \omega$. Altogether at most one summand of type (M) is nonzero, and one is if and only if C is $u\rho$-tight and either $D=\{k-1\}$, or D is nontrivial with $\max D = k-1$ and ω-tight. This is exactly the condition for $u \overset{k-2}{\rightsquigarrow} w$: $\pi=\rho CD$ is $(k-2)$-shaped (as $k-1\notin C$, C has the unique element k greater than $k-2$, and D the unique element $k-1$) and it is u-tight if and only if C is $u\rho$-tight and D is ω-tight. When it holds, Lemma 6.19(c) gives

$$\kappa_{ak} q_{k-1}(u,wt_{ak}) = \mathbf{q}_u(\rho)\mathbf{q}_u(C)\,\kappa_{ak}\mathbf{q}_\omega(\tau_a) = \mathbf{q}_u(\rho)\mathbf{q}_u(C)\,q_{k-1}\mathbf{q}_\omega(D) = q_{k-1}q_{k-2}(u,w),$$

and $\mathrm{fix}_{k-1}(u,wt_{ak}) = \mathrm{fix}_{k-1}(u,w)\setminus\{u(a)\} = \mathrm{fix}_{k-2}(u,w)$ as in the proof of Lemma 6.20. Thus the summand of type (M) equals $-q_{k-1}c_{k-2}(u,w)$ and cancels the quantum term; when the condition fails, both vanish. □

Lemma 6.22 (Vanishing)**.** *Suppose $u,w\in S_\infty$, let $k>1$, and suppose $u \overset{k}{\not\rightsquigarrow} w$. Then*

$$(y_{w(k)}-z_1)\,c_{k-1}(u,w) + \sum_{b>k,\ wt_{kb}\lhd w} \kappa_{kb}\,c_{k-1}(u,wt_{kb}) - \sum_{a<k,\ wt_{ak}\lhd w} \kappa_{ak}\,c_{k-1}(u,wt_{ak}) + q_{k-1}c_{k-2}(u,w) = 0,$$

where $\kappa_{rs}=\kappa(wt_{rs},t_{rs})$.

Proof. If $u(k)=w(k)$, then $u \overset{k-1}{\not\rightsquigarrow} w$ by Lemma 6.14(i), so the diagonal term vanishes, and the rest is Lemma 6.20. If $u(k)\neq w(k)$ and $B>k$, the diagonal and quantum terms vanish and the Monk summands sum to $c_k(u,w)=0$ by Lemma 6.21(a). If $u(k)\neq w(k)$ and $B=k$, this is Lemma 6.21(b). □

We now record the recurrence satisfied by the quantum factorial elementary polynomials, which drives the induction.

Lemma 6.23. *For all integers k we have*

$$\widetilde{E}_{k,k}(x;z) = (x_k - z_1)\widetilde{E}_{k-1,k-1}(x;z) + q_{k-1}\widetilde{E}_{k-2,k-2}(x;z),$$

with the conventions $\widetilde{E}_{0,0}=1$ and $\widetilde{E}_{k,k}=0$ if $k<0$. Moreover, the ordinary polynomials $\mathbf{E}(y_A;z)=\prod_{i\in A}(y_i-z_1)$ satisfy, for any set A and any $c\notin A$, the single-variable identity

$$\mathbf{E}(y_{A\cup\{c\}};z) = (y_c - z_1)\,\mathbf{E}(y_A;z).$$

Proof. By Lemma 3.3, $\widetilde{E}_{k,k}(x;z)=\Delta_k(-z_1 \mid x_1,\dots,x_k)$, and expanding the tridiagonal determinant $\Delta_k(t\mid x_1,\dots,x_k)$ along its last row gives $\Delta_k=(x_k+t)\Delta_{k-1}+q_{k-1}\Delta_{k-2}$. The second identity is clear. □

The first recurrence is the quantization of the classical relation $E_{k,k}=(x_k-z_1)E_{k-1,k-1}$; the extra term $q_{k-1}\widetilde{E}_{k-2,k-2}$ is the only feature absent from the non-quantum setting. Discarding all quantum terms recovers the classical Pieri formula [16, Theorem 7.1], whose term-by-term cancellation is carried out in [16, Proof of Proposition 4.4] by means of the reflection-matching Lemmas 7.6–7.9 of that paper; the lemmas above are the quantum counterparts, the essential new points being the length-decreasing covers permitted by Lemma 6.8, the attendant bookkeeping of the powers of q, and the extra term of the recurrence, which is absorbed by the merge summands of Lemma 6.19.

Proof of Theorem 3.4. For a permutation w, write $[\widetilde{\mathfrak{S}}_w]F$ for the coefficient of $\widetilde{\mathfrak{S}}_w(x;y)$ in an expansion F of a quantum double Schubert polynomial in the basis $\{\widetilde{\mathfrak{S}}_w(x;y)\}$. With this notation the theorem is the assertion that, for all u and all $k\geq 1$,

$$[\widetilde{\mathfrak{S}}_w]\big(\,\widetilde{\mathfrak{S}}_u(x;y)\widetilde{E}_{k,k}(x;z)\big) = \begin{cases} q_k(u,w)\,\mathbf{E}(y_{\mathrm{fix}_k(u,w)};z) & \text{if } u \overset{k}{\rightsquigarrow} w,\\ 0 & \text{otherwise.}\end{cases}$$

For brevity, when $u \overset{m}{\rightsquigarrow} w$ we set

$$c_m(u,w) = q_m(u,w)\,\mathbf{E}(y_{\mathrm{fix}_m(u,w)};z),$$

and $c_m(u,w)=0$ if $u \overset{m}{\not\rightsquigarrow} w$; note that $u \overset{0}{\rightsquigarrow} w$ only for $w=u$, with $c_0(u,u)=1$, in accordance with $\widetilde{E}_{0,0}=1$. The claim is that $[\widetilde{\mathfrak{S}}_w](\widetilde{\mathfrak{S}}_u\,\widetilde{E}_{k,k}) = c_k(u,w)$.

We argue by strong induction on k.

Base case $k=1$. We have $\widetilde{E}_{1,1}(x;z)=x_1-z_1$, and Lemma 6.1 with $i=j=1$ gives

$$(x_1-z_1)\,\widetilde{\mathfrak{S}}_u(x;y) = (y_{u(1)}-z_1)\,\widetilde{\mathfrak{S}}_u(x;y) + \sum_{\substack{b>1\\ \ell(ut_{1b})=\ell(u)+1}} \widetilde{\mathfrak{S}}_{ut_{1b}}(x;y) + \sum_{\substack{b>1\\ \ell(ut_{1b})=\ell(u)-2(b-1)+1}} q_{1b}\,\widetilde{\mathfrak{S}}_{ut_{1b}}(x;y).$$

The permutations w with $u \overset{1}{\rightsquigarrow} w$ are exactly $w=u$ and $w=ut_{1b}$ for $b>1$ with $u \lhd ut_{1b}$, since Definition 3.1 permits at most one reflection when $k=1$. For $w=u$ we have $q_1=1$, $\mathrm{fix}_1(u,u)=\{u(1)\}$, and $\mathbf{E}(y_{\{u(1)\}};z)=y_{u(1)}-z_1$, matching the diagonal term. For $w=ut_{1b}$ we have $\mathrm{fix}_1=\emptyset$ and $\mathbf{E}(y_\emptyset;z)=1$; if the step is length-increasing then $q_1(u,w)=1$, matching the second sum, and if it is length-decreasing then $q_1(u,w)=q_{1b}$, matching the third sum. All other coefficients vanish on both sides. This proves the base case.

Inductive step. Fix $k>1$ and assume the claim at all levels $m<k$. By Lemma 6.23,

$$\widetilde{\mathfrak{S}}_u\,\widetilde{E}_{k,k} = (x_k-z_1)\big(\widetilde{\mathfrak{S}}_u\,\widetilde{E}_{k-1,k-1}\big) + q_{k-1}\,\widetilde{\mathfrak{S}}_u\,\widetilde{E}_{k-2,k-2}. \tag{6.6}$$

By the inductive hypothesis, $\widetilde{\mathfrak{S}}_u\,\widetilde{E}_{k-1,k-1} = \sum_v c_{k-1}(u,v)\,\widetilde{\mathfrak{S}}_v$. Applying Lemma 6.1 with $i=k$, $j=1$ to each $\widetilde{\mathfrak{S}}_v$ and extracting the coefficient of $\widetilde{\mathfrak{S}}_w$ from (6.6), we obtain

$$[\widetilde{\mathfrak{S}}_w]\big(\widetilde{\mathfrak{S}}_u\,\widetilde{E}_{k,k}\big) = (y_{w(k)}-z_1)\,c_{k-1}(u,w) \tag{6.7}$$

$$+ \sum_{\substack{b>k,\ wt_{kb}\lhd w\\ \ell(w)=\ell(wt_{kb})+1}} c_{k-1}(u,wt_{kb}) + \sum_{\substack{b>k,\ wt_{kb}\lhd w\\ \ell(w)=\ell(wt_{kb})-2(b-k)+1}} q_{kb}\,c_{k-1}(u,wt_{kb}) \tag{6.8}$$

$$- \sum_{\substack{a<k,\ wt_{ak}\lhd w\\ \ell(w)=\ell(wt_{ak})+1}} c_{k-1}(u,wt_{ak}) - \sum_{\substack{a<k,\ wt_{ak}\lhd w\\ \ell(w)=\ell(wt_{ak})-2(k-a)+1}} q_{ak}\,c_{k-1}(u,wt_{ak}) \tag{6.9}$$

$$+ q_{k-1}\,c_{k-2}(u,w), \tag{6.10}$$

where we used that $\mathrm{sign}(b-k)=1$ for $b>k$ and $\mathrm{sign}(a-k)=-1$ for $a<k$, that the term $\widetilde{\mathfrak{S}}_w$ arises from $\widetilde{\mathfrak{S}}_v$ with $v=wt$ exactly when $v \lhd w$, and that a summand in (6.8)–(6.9) is present precisely when $u \overset{k-1}{\rightsquigarrow} wt_{kb}$, respectively $u \overset{k-1}{\rightsquigarrow} wt_{ak}$ (otherwise $c_{k-1}=0$). In the terminology introduced before Lemma 6.15, (6.8) and (6.9) are the Monk summands at w, with κ_{kb}, respectively κ_{ak}, equal to 1 in the first sum and to q_{kb}, respectively q_{ak}, in the second. We must show that (6.7)–(6.10) equals $c_k(u,w)$. We distinguish three cases according to whether $u(k)=w(k)$ and, when $u(k)\neq w(k)$, whether the cycle C of $u^{-1}w$ containing k has its top B greater than k or equal to k.

Case 1: $u(k)=w(k)$. By Lemma 6.14(i) we have $u \overset{k}{\rightsquigarrow} w$ if and only if $u \overset{k-1}{\rightsquigarrow} w$; suppose first that both hold. Then, by the same lemma,

$$\mathrm{fix}_k(u,w) = \mathrm{fix}_{k-1}(u,w)\cup\{u(k)\}, \qquad q_k(u,w)=q_{k-1}(u,w),$$

so the diagonal contribution (6.7) gives

$$(y_{u(k)}-z_1)\,c_{k-1}(u,w) = q_{k-1}(u,w)\,(y_{u(k)}-z_1)\,\mathbf{E}(y_{\mathrm{fix}_{k-1}(u,w)};z) = q_k(u,w)\,\mathbf{E}(y_{\mathrm{fix}_k(u,w)};z) = c_k(u,w),$$

by Lemma 6.23. The remaining contributions are the reflection sums (6.8), (6.9) and the quantum term $q_{k-1}c_{k-2}(u,w)$ of (6.10), which sum to 0 by Lemma 6.20. Hence the total is $c_k(u,w)$, as required. If instead $u(k)=w(k)$ but $u \overset{k}{\not\rightsquigarrow} w$ (equivalently $u \overset{k-1}{\not\rightsquigarrow} w$), then $c_k(u,w)=0$ and the diagonal term vanishes, while the reflection sums and quantum term again cancel by Lemma 6.20; the total is $0=c_k(u,w)$.

Case 2: $u(k)\neq w(k)$ and $B>k$. By Lemma 6.21(a), the diagonal term (6.7) and the quantum term (6.10) vanish, and the Monk summands (6.8)–(6.9) sum to $c_k(u,w)$. (When $u \overset{k}{\rightsquigarrow} w$, Lemma 6.17 identifies the single surviving summand: it is indexed by t_{kB}, with $v=wt_{kB}$, $u \overset{k-1}{\rightsquigarrow} v$, $\mathrm{fix}_k(u,w)=\mathrm{fix}_{k-1}(u,v)$ and $\kappa_{kB}\,q_{k-1}(u,v)=q_k(u,w)$, the factor κ_{kB} being 1 or q_{kB} according as t_{kB} is a classical or a quantum cover

of v; this is the step at which the quantum drop terms of Lemma 6.1 and the quantum weight of Definition 3.1 interlock.)

Case 3: $u(k) \neq w(k)$ and $B = k$. Here $u \overset{k}{\not\rightsquigarrow} w$, since C contains no element greater than k, so $c_k(u,w) = 0$; and (6.7)–(6.10) vanishes by Lemma 6.21(b) (equivalently, by Lemma 6.22).

By Lemma 6.14 these cases are exhaustive. In all cases $[\widetilde{\mathfrak{S}}_w](\widetilde{\mathfrak{S}}_u\, \widetilde{E}_{k,k}) = c_k(u,w)$, completing the induction and the proof. □

7. The iterated Pieri formula

Iterating Theorem 3.4 gives the expansion of a product with two quantum factorial elementary polynomials in different alphabets.

Corollary 7.1 (Iterated quantum Pieri formula)**.** *For all $u \in S_\infty$ and $k \geq 1$,*

$$\widetilde{\mathfrak{S}}_u(x;y)\widetilde{E}_{k,k}(x;z_1)\widetilde{E}_{k,k}(x;z_2) = \sum_{u \overset{k}{\rightsquigarrow} v \overset{k}{\rightsquigarrow} w} q_k(u,v)q_k(v,w) \prod_{a \in \mathrm{fix}_k(u,v)} (y_a - z_1) \prod_{b \in \mathrm{fix}_k(v,w)} (y_b - z_2)\, \widetilde{\mathfrak{S}}_w(x;y),$$

where $\widetilde{E}_{k,k}(x;z_i)$ denotes $\widetilde{E}_{k,k}(x;z)$ with z_1 replaced by z_i.

Proof. Apply Theorem 3.4 to multiply $\widetilde{\mathfrak{S}}_u(x;y)$ by $\widetilde{E}_{k,k}(x;z_1)$. For each resulting term indexed by v, apply the same theorem with the z-alphabet replaced by z_2. The coefficient factors multiply, and the two successive relations are exactly $u \overset{k}{\rightsquigarrow} v \overset{k}{\rightsquigarrow} w$. □

Remark 7.2 (Two-column shapes). Let $d_2[k]$ be the dominant permutation with

$$\mathfrak{c}(d_2[k]) = \overbrace{(2,2,\ldots,2)}^{k \text{ times}},$$

whose diagram is a $k \times 2$ rectangle. Classically $\mathfrak{S}_{d_2[k]}(x;z) = E_{k,k}(x;z_1)E_{k,k}(x;z_2)$ is a product of two factorial elementary polynomials, so Corollary 7.1 at $q = 0$ is a Pieri rule for multiplication by $\mathfrak{S}_{d_2[k]}(x;z)$. In the quantum setting, however, $\widetilde{\mathfrak{S}}_{d_2[k]}(x;z)$ is *not* the product $\widetilde{E}_{k,k}(x;z_1)\widetilde{E}_{k,k}(x;z_2)$: the two columns of the rectangle have the same height, so $x^{(2^k)}$ is not a standard elementary monomial, and the Fomin–Gelfand–Postnikov quantization does not factor. Instead one has a Giambelli-type determinant. For a Grassmannian permutation w with descent at k and shape λ (so $\mathfrak{c}(w) = (\lambda_k, \ldots, \lambda_1)$), the double Schubert polynomial is the determinant $\mathfrak{S}_w(x;z) = \det\left(E_{\lambda_i'+j-i,\, k+j-1}(x;z)\right)_{1 \leq i,j \leq \lambda_1}$ of factorial elementary polynomials in the nested alphabets $x_1, \ldots, x_{k+j-1}$; every term of the expanded determinant is a standard elementary monomial, so the quantization map acts on it term by term and gives the quantum Giambelli formula [5, 3]

$$\widetilde{\mathfrak{S}}_w(x;z) = \det\left(\widetilde{E}_{\lambda_i'+j-i,\, k+j-1}(x;z)\right)_{1 \leq i,j \leq \lambda_1}$$

in terms of the polynomials of Definition 3.2. For the $k \times 2$ rectangle this reads

$$\widetilde{\mathfrak{S}}_{d_2[k]}(x;z) = \widetilde{E}_{k,k+1}(x;z)\, \widetilde{E}_{k,k}(x;z) - \widetilde{E}_{k+1,k+1}(x;z)\, \widetilde{E}_{k-1,k}(x;z),$$

whose first instances are

$$\widetilde{\mathfrak{S}}_{d_2[1]}(x;z) = \widetilde{E}_{1,2}(x;z)\widetilde{E}_{1,1}(x;z) - \widetilde{E}_{2,2}(x;z), \qquad \widetilde{\mathfrak{S}}_{d_2[2]}(x;z) = \widetilde{E}_{2,3}(x;z)\widetilde{E}_{2,2}(x;z) - \widetilde{E}_{3,3}(x;z)\widetilde{E}_{1,2}(x;z).$$

Consequently a product $\widetilde{\mathfrak{S}}_u(x;y)\, \widetilde{\mathfrak{S}}_{d_2[k]}(x;z)$ is computed by two successive quantum Pieri steps (for $\widetilde{E}_{k,k}$ and $\widetilde{E}_{k,k+1}$) minus two others; this is the "paired step" of the quantum multiplication algorithm of Section 4, and it is the reason that algorithm iterates over the medium theta $\overline{\theta}(v^{-1})$ rather than over $\theta(v^{-1})$. We do not know a closed Pieri-type formula for $\widetilde{\mathfrak{S}}_u(x;y)\, \widetilde{\mathfrak{S}}_{d_2[k]}(x;z)$ whose coefficients are individually positive.

8. The Peterson–Woodward comparison formula

In this section we prove Proposition 2.10. We use the notation of Section 2: W_P is a parabolic subgroup of S_n with $P = \widehat{P}$, $\omega_P = w_P$ is its longest element, and for $\lambda_P \in Q^\vee/Q_P^\vee$ we write $\lambda_B \in Q^\vee$ for its Peterson–Woodward lift. In the language of roots, λ_B is the unique lift with $\langle \alpha, \lambda_B \rangle \in \{0, -1\}$ for all positive roots α of the Levi subgroup of P; in type A these are the roots $\alpha_i + \cdots + \alpha_{j-1}$ with $s_i, \ldots, s_{j-1} \in P$, and $\langle \alpha_i + \cdots + \alpha_{j-1}, d \rangle = \langle d \rangle_{ij}$, so this is the condition of Definition 2.8.

Proposition 8.1 ([7, Proposition 2.10])**.** *Let W_P be a parabolic subgroup of W and $u, v, w \in W^P$. Then for any $\lambda_P \in Q^\vee/Q_P^\vee$, we have that*

$$c_{u,v}^{w,\lambda_P}(y;y) = c_{u,v}^{w\omega_P\omega_{P'},\lambda_B}(y;y),$$

where ω_P is the longest element in W_P, $\omega_{P'}$ is the longest element in $W_{P'}$, where P' is the parabolic subgroup generated by $\{s_\alpha \mid \alpha \in \Delta_P, \langle \alpha, \lambda_B\rangle = 0\}$, and λ_B is the Peterson-Woodward lift of λ_P.

Here the left-hand side is a structure constant of $QH_T^*(G/P)$, i.e. of the parabolic quantum double Schubert polynomials $\widetilde{\mathfrak{S}}_w^P(x;y)$ of Definition 2.6, which represent the Schubert classes of $QH_T^*(G/P)$ [10], and the right-hand side is a structure constant of $QH_T^*(G/B)$.

Corollary 8.2. *With the notation of Proposition 8.1, for all $u, v, w \in W^P$,*

$$c_{u,v}^{w,\lambda_P}(y;z) = c_{u,v}^{w\omega_P\omega_{P'},\lambda_B}(y;z).$$

Proof. The parabolic polynomials satisfy the same quantum Cauchy identity as in Lemma 2.1. Indeed, by [10, Lemma 18], $\widetilde{\mathfrak{S}}_w^P(x;y) = \theta^P(\mathfrak{S}_w(x;y))$ for $w \in W^P$, where θ^P is the parabolic quantization map of Ciocan-Fontanine [3], extended linearly over the coefficient variables; applying θ^P to the classical identity $\mathfrak{S}_v(x;z) = \sum_{v_1\cdot v_2 = v} \mathfrak{S}_{v_1}(y;z)\,\mathfrak{S}_{v_2}(x;y)$ gives $\widetilde{\mathfrak{S}}_v^P(x;z) = \sum_{v_1\cdot v_2=v} \mathfrak{S}_{v_1}(y;z)\,\widetilde{\mathfrak{S}}_{v_2}^P(x;y)$, where we note that $v_2 \in W^P$ for every reduced factorization (see below), so that $\theta^P(\mathfrak{S}_{v_2}(x;y))$ is indeed $\widetilde{\mathfrak{S}}_{v_2}^P(x;y)$. Hence

$$c_{u,v}^{w,\lambda_P}(y;z) = \sum_{v_1\cdot v_2 = v} c_{u,v_2}^{w,\lambda_P}(y;y)\ \mathfrak{S}_{v_1}(y;z).$$

Here, if $v = v_1 \cdot v_2$ is reduced and $v \in W^P$, then $v_2 \in W^P$: for $s_i \in P$, $v_2 s_i < v_2$ would give $\ell(vs_i) \le \ell(v_1) + \ell(v_2 s_i) < \ell(v)$, contradicting $v \in W^P$. Thus Proposition 8.1 applies to each term, and resumming by Lemma 2.1 gives

$$\sum_{v_1\cdot v_2 = v} c_{u,v_2}^{w\omega_P\omega_{P'},\lambda_B}(y;y)\ \mathfrak{S}_{v_1}(y;z) = c_{u,v}^{w\omega_P\omega_{P'},\lambda_B}(y;z). \quad \square$$

Proof of Proposition 2.10. Let $d_P \in Q^\vee/Q_P^\vee$ with Peterson–Woodward lift $d = \lambda_B$. In type A, $\langle \alpha_i, d\rangle = 2d_i - d_{i-1} - d_{i+1} = \langle d\rangle_i$, so the parabolic subgroup P' of Proposition 8.1 is generated by $\{s_i \in P \mid \langle d\rangle_i = 0\}$, that is, by $\{s_i \mid i \in z_P(d)\}$, and $\omega_P\omega_{P'} = w_P w_{z_P(d)}$. The claim is therefore Corollary 8.2. (The passage from S_n to S_∞, i.e. from P to $\widehat{P}$, is the stability discussed after Definition 2.6.) $\square$

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